

Saturated heterocycles and stereoelectronics

31

Connections

➡ Building on

- Acetals and hemiacetals [ch11](#)
- Stereochemistry [ch13](#)
- The conformation of cyclic molecules [ch16](#)
- Stereospecific elimination reactions [ch17](#)
- Proton NMR [ch18](#)
- Aldol reactions [ch26](#)
- Aromatic heterocycles [ch29](#) & [ch30](#)

Arriving at

- Putting a heteroatom in a ring changes the reactivity of the heteroatom
- Ring-opening reactions: the effect of ring strain
- Lone pairs in heterocycles have precise orientations
- Some substituents prefer to be axial on some six-membered saturated heterocycles
- Interactions of lone pairs with empty orbitals can control conformation
- Ring-closing reactions: why five-membered rings form quickly and four-membered rings form slowly
- Baldwin's rules: why some ring closures work well while others don't work at all
- How conformation and ring size affect coupling constants
- Geminal coupling
- The relationship between symmetry and NMR spectra: diastereotopicity

➡ Looking forward to

- Stereoselectivity in cyclic systems [ch32](#)
- Diastereoselectivity [ch33](#)
- Asymmetric synthesis [ch41](#)
- Chemistry of life [ch42](#)

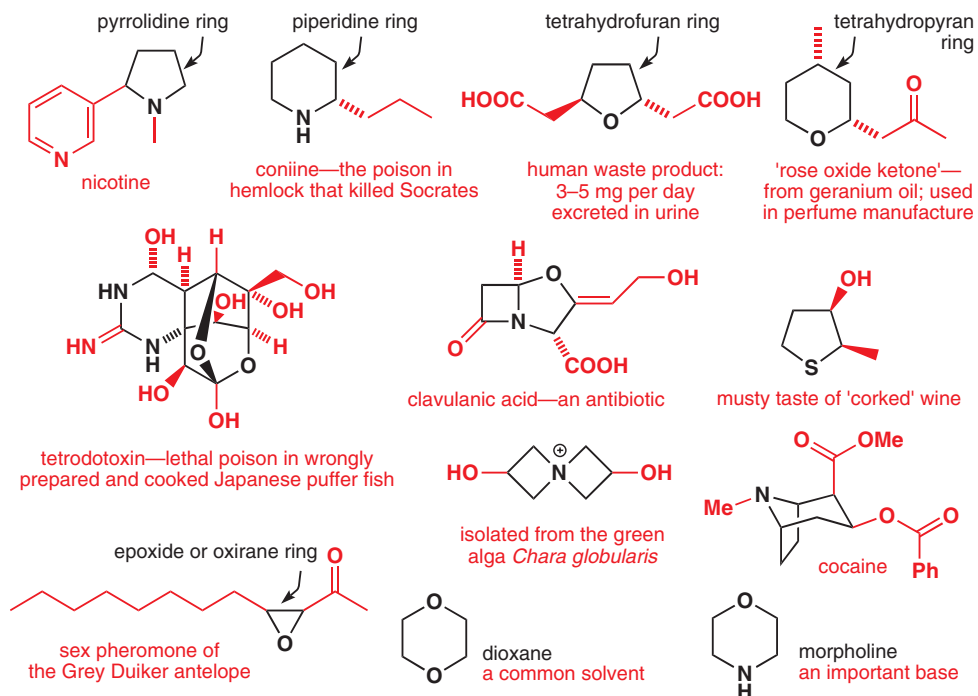
Introduction

Rings make a difference to the way molecules react and the ways they can be made, and we have just devoted two chapters to the reactions and synthesis of flat, aromatic heterocycles. In this chapter and the one that follows we shall continue to look at rings, but not flat aromatic ones. Once you put saturated atoms into rings the rings become flexible and display interesting chemical features. We introduced ways of talking about conformation in rings in Chapter 16, and we will revisit ideas from that chapter—in particular we will build on the idea that rings make it easier to think about stereochemistry because they restrict the number of conformations a molecule can adopt. We will also introduce a theme which we develop over the next few chapters of the book: *stereoselectivity*—how to make single diastereoisomers of a product.

It may seem strange that heterocycles—rings containing not just carbon atoms, but oxygen, nitrogen, or sulfur as well—deserve three whole chapters, but you will soon see that this is justified both by the sheer number and variety of heterocycles that exist and by their special chemical features. We dealt with the special stereochemical features of *aromatic* heterocycles

Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.

in the last two chapters, in particular their distinctive reactivity, stability, and ease of synthesis. Some examples of *saturated* heterocycles, a few of which may be familiar to you, are shown below.



The saturated heterocyclic rings are shown in black, and names for the most important ring types are given: some (like piperidine, morpholine) you will need to remember; others (tetrahydrofuran, pyrrolidine) are more obviously derived from the names for aromatic heterocycles you met in the last chapter. Some of these compounds (nicotine, coniine, cocaine) are plant products falling into the class called alkaloids, which are discussed in Chapter 42. Another important class of saturated heterocycles, sugars, will also appear in Chapter 42.

But what are the 'special chemical features' of saturated heterocycles? Putting a heteroatom into a ring does two important things, and these lead to the most important new topics in this chapter.

- Firstly, the heteroatom makes the ring easy to make by a ring-closing reaction, or (in some cases) easy to break by a ring-opening reaction. Closing and opening reactions of rings are subject to constraints that you will need to know about, and the principles that govern these reactions are discussed later in the chapter.
- Secondly, the ring fixes the orientation of the heteroatom—and, in particular, the orientation of its lone pairs—relative to the atoms around it. This has consequences for the reactivity and conformation of the heterocycle which can be explained using the concept of stereoelectronics.

• Stereoelectronic effects are chemical consequences of the arrangement of orbitals in space.

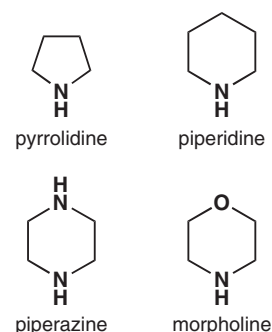
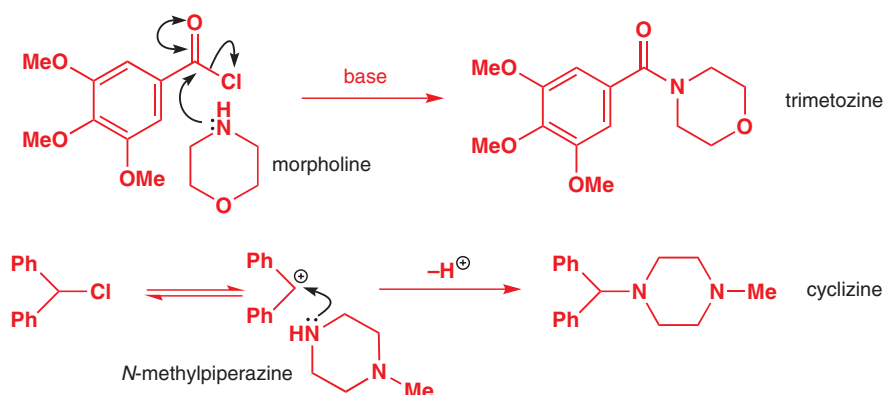
Reactions of saturated heterocycles

Saturated nitrogen heterocycles: amines, but more nucleophilic

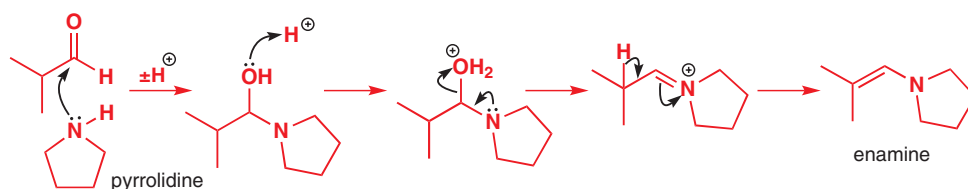
In many reactions the simple saturated nitrogen heterocycles—piperidine, pyrrolidine, piperazine, and morpholine—behave simply as secondary amines that happen to be cyclic.

■ Although this is the only chapter in which stereoelectronics appears in the title, you will soon recognize the similarity between the ideas we cover here and concepts like the stereospecificity of E2 elimination reactions (Chapter 17) and the effect of orbital overlap on NMR coupling constants (Chapter 18). We will also use orbital alignment to explain the Karplus relationship (Chapter 32), the Felkin–Anh transition state (Chapter 33), and the conformational requirements for rearrangement and fragmentation reactions (Chapter 36).

They do the sorts of things that other amines do, acting as nucleophiles in addition and substitution reactions. Morpholine, for example, is acylated by 3,4,5-trimethoxybenzoyl chloride to form the tranquilizer and muscle relaxant trimetozine, and *N*-methylpiperazine can be alkylated in an S_N1 reaction with diphenylmethyl chloride to give the travel-sickness drug cyclizine.



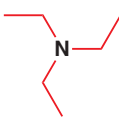
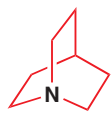
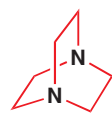
The addition of pyrrolidine to aldehydes and ketones is a particularly important reaction because it leads to enamines, the valuable enol equivalents discussed in Chapter 25.



Enamines formed from pyrrolidine and piperidine are particularly stable because pyrrolidine and piperidine are rather more nucleophilic than comparable acyclic amines such as diethylamine. This is a general feature of cyclic amines (and cyclic ethers, too, as you will see shortly), and is a steric effect. The alkyl substituents, being tied back into a ring, are held clear of the nucleophilic lone pair, allowing it to approach an electrophile without hindrance. This effect is well illustrated by comparing the rates of reaction of methyl iodide with three amines—tertiary this time. The two cyclic compounds are bridged—quinuclidine is a bridged piperidine while the diamine known as DABCO (1,4-diazabicyclo[2.2.2]octane) is a bridged piperazine. The table below shows the relative rates, along with pK_a values, for triethylamine, quinuclidine, and DABCO.

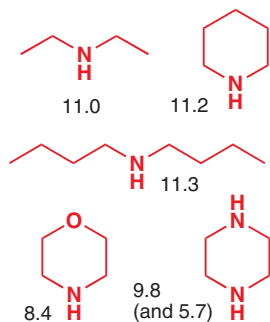


Rates of reaction of amines with methyl iodide

			
	triethylamine	quinuclidine	DABCO
relative rate of reaction ^a	1	63	40
pK_a of R_3NH^+	10.7	11.0	8.8 (and 3.0)

^aRelative rate of reaction with MeI in MeCN at 20°C.

pK_a of $R_2NH_2^+$
for some secondary amines



■ To clarify: the pK_a s we are talking about here are the pK_a values of the ammonium ions $R_2NH_2^+$.

Quinuclidine and DABCO are 40–60 times more reactive than triethylamine. This is again due to the way the ring structures keep the nitrogen's substituents away from interfering with the lone pair as it attacks the electrophile. You should contrast the effect that the cyclic structure has on the basicity of the amines: none! Triethylamine and quinuclidine are equally basic and, as you can see in the margin, so (more or less) are diethylamine, dibutylamine, and piperidine. A proton is so small that it cares very little whether the alkyl groups are tied back or not.

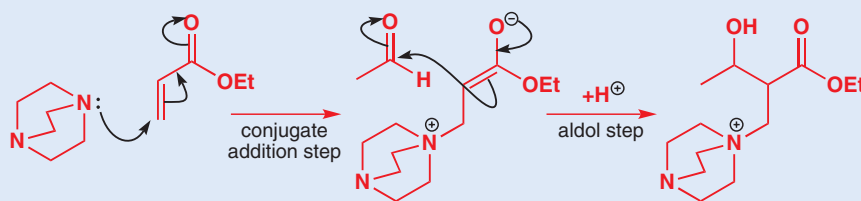
Much more important in determining pK_a is how electron-rich the nitrogen is, and this is the cause of the glaring discrepancy between the basicity of quinuclidine and that of DABCO, or between the basicities of piperidine (pK_a 11.2) and morpholine (pK_a 9.8) or piperazine (pK_a 8.4). The extra heteroatom, through an inductive effect, withdraws electron density from the nitrogen atom, making it less nucleophilic and less basic. In this sense, morpholine can be a very useful base, less basic than triethylamine but somewhat more so than pyridine (pK_a 5.2). Notice how much lower is the second pK_a (that is, the pK_a for protonation of the second nitrogen) of the diamines DABCO and piperazine: the protonated nitrogen of the monoprotonated amine withdraws electrons very effectively from the unprotonated one.

The Baylis–Hillman reaction

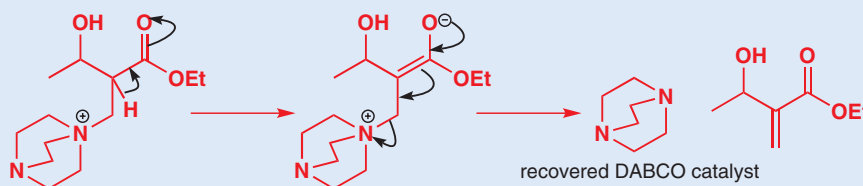
One of the most important uses of DABCO is in the Baylis–Hillman reaction, discovered in 1972 by two chemists at the Celanese Corporation in New York. Their reaction is a modification of the aldol reaction (Chapter 26), except that instead of the enolate being formed by deprotonation it is formed by conjugate addition. You have seen the enolate products of conjugate addition being trapped by alkylating agents in Chapter 25, but in the Baylis–Hillman reaction the electrophile is an aldehyde and is present right from the start of the reaction, which is done just by stirring the components at room temperature. Here is a typical example.



The reaction starts with the (relatively nucleophilic) DABCO undergoing conjugate addition to ethyl acrylate. This will form an enolate that can then attack the acetaldehyde in an aldol reaction.

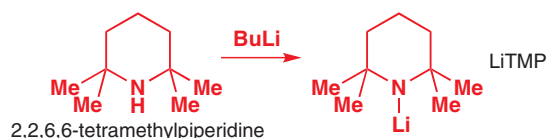


E1cB eliminations often follow aldol reactions and lead to α,β -unsaturated products. In this case, though, DABCO is a much better leaving group than the hydroxyl group, so enolization leads to loss of DABCO in an E1cB elimination, giving the product of the reaction. DABCO is recovered and is a catalyst.

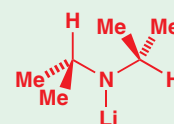


A disadvantage of the Baylis–Hillman reaction is its rate: typically, several days' reaction time are required. Pressure helps speed the reaction up, but as a catalyst DABCO is about the best. It is nucleophilic because of the 'tied back' alkyl groups, but importantly it is a good leaving group because it has a relatively low pK_a , meaning that it leaves easily in the last step. As you have seen before, good nucleophiles are usually bad leaving groups, although there are many exceptions. DABCO's combination of nucleophilicity and leaving group ability is perfect here.

The exposed nature of the nitrogen atom in cyclic amines means that nitrogen heterocycles are very frequently encountered in drug molecules, particularly those operating on the central nervous system (cocaine, heroin, and morphine all contain nitrogen heterocycles, as do codeine and many tranquilizers, such as Valium). But the ring can also be used as a support for adding substituents that hinder the nitrogen's lone pair. Just as the nitrogen atom of piperidine is permanently exposed, the nitrogen atom of 2,2,6,6-tetramethylpiperidine (TMP) nestles deep in a bed of methyl groups. The lithium salt of TMP (LiTMP) is an analogue of LDA—a base that experiences enormous steric hindrance that can be used in situations where the selectivity even of LDA fails.

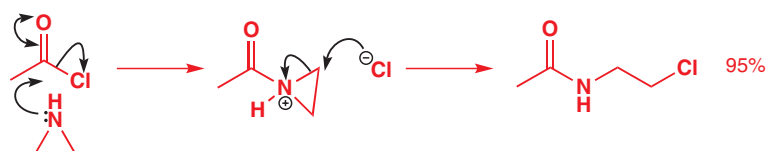


■ With LDA, one or other of the isopropyl groups always has the option of rotating to place only a C–H group close to the N–Li bond. In LiTMP, there are unavoidably four Me groups close to Li.

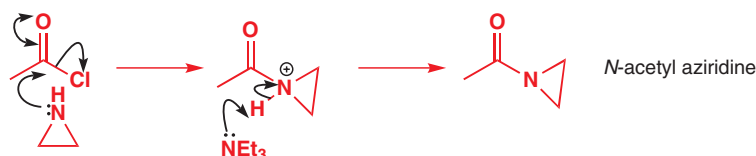


Aziridine: ring strain promotes ring opening

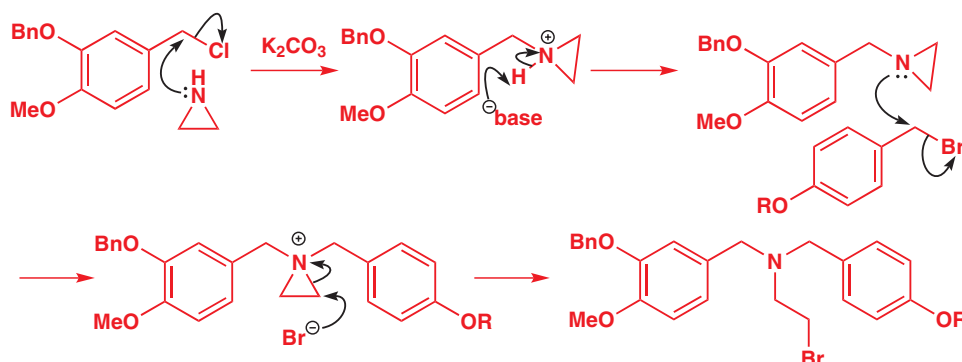
Aziridine and azetidine are stable, if volatile, members of the saturated nitrogen heterocycle family, and aziridine has some interesting chemistry of its own. Like pyrrolidine and piperidine, aziridine can be acylated by treatment with an acyl chloride, but the product is not stable. The ring opens with attack of chloride, a relatively poor nucleophile, and an open-chain secondary amide results.



You can view this ring opening as very similar to the ring opening of an epoxide (Chapter 19)—in particular, a *protonated* epoxide, in which the oxygen bears a positive charge. The positive charge is very important for aziridine opening because, when the reaction is done in the presence of a base, removal of the proton leads immediately to the neutral acyl aziridine, which is stable.



The ring opening of aziridine is a useful way of making larger heterocycles: anything that puts a positive charge on nitrogen encourages the opening by making N a better leaving group, whether it's protonation or, as shown below, alkylation. Alkylation of aziridine in base gives the *N*-substituted aziridine as you might expect, but a second alkylation leads to a positively charged aziridinium salt that opens immediately to a useful bromoamine.



Systematic nomenclature of saturated heterocycles

The names aziridine and azetidine are derived from a reasonably logical system of nomenclature, which assigns three-part heterocycle names according to: (a) the heteroatom ('az-' = nitrogen, 'ox-' = oxygen, 'thi-' = sulfur), (b) the ring size ('-ir-' = 3, from tri; '-et-' = 4, from tetra; '-ol-' = 5; nothing for 6; '-ep-' = 7, from hepta; '-oc-' = 8, from octa; etc.), and (c) the degree of saturation ('-ene' or '-ine' for unsaturated, '-idine' or '-ane' for saturated). Hence az-ir-idine, az-et-idine, di-ox-ol-ane, and ox-ir-ane.

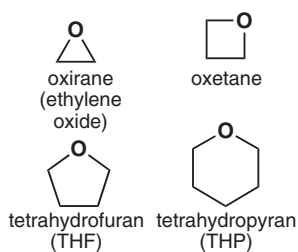
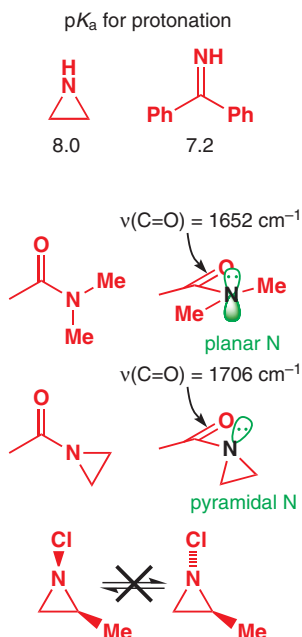
■ In this case, the product is an intermediate in the synthesis of two natural products, sandaverine and corgoine.

We have just mentioned the protonation of aziridine, and you might imagine from what we said earlier about the comparative nucleophilicity and basicity of nitrogen heterocycles and their acyclic counterparts that aziridine will be even more nucleophilic than pyrrolidine, and about as basic. Well, it isn't. The idea that 'tying back' the alkyl groups increases nucleophilicity is only valid for unstrained five or six-membered rings: with small rings another effect takes over.

Aziridine is, in fact, much less basic than pyrrolidine and piperidine: the pK_a for its protonation is only 8.0. This is much closer to the pK_a of a compound containing an sp^2 hybridized nitrogen atom—the imine in the margin, for example. This is because the nitrogen's lone pair is in an orbital with much more s character than is typical for an amine, due to the three-membered ring. This is an effect we have discussed before, in Chapter 18, and you should re-read pp. 412–415 if you need to refresh your memory. There we compared three-membered rings with alkynes, explaining that both could be deprotonated relatively easily. The anion carries a negative charge in a low-energy orbital with much s character: the same type of orbital carries aziridine's lone pair.

The s character of the aziridine nitrogen's lone pair has other effects too. The lone pair interacts very poorly with an adjacent carbonyl group, so N -acyl aziridines such as the one you saw on p. 973 behave not at all like amides. The nitrogen atom is pyramidal and not planar, and the stretching frequency of the $C=O$ bond (1706 cm^{-1}) is much closer to that of a ketone (1710 cm^{-1}) than that of an amide (1650 cm^{-1}).

The s character of the lone pair means that the nitrogen atom inverts very slowly, rather like a phosphine. Usually it is not possible for nitrogen to be a stereogenic centre because inversion is too rapid—the transition state for nitrogen inversions (in which the lone pair is in a p orbital) is low in energy. But with an aziridine, getting the lone pair into a p orbital requires much more activation energy, so nitrogen can be stereogenic. The two stereoisomers of the N -substituted aziridine in the margin can be separated and isolated.

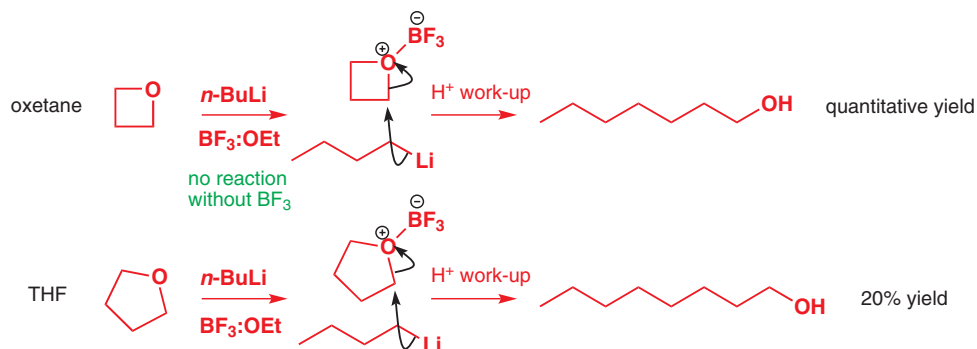


➡ Epoxide opening under acidic and basic conditions was covered in Chapter 19.

Oxygen heterocycles

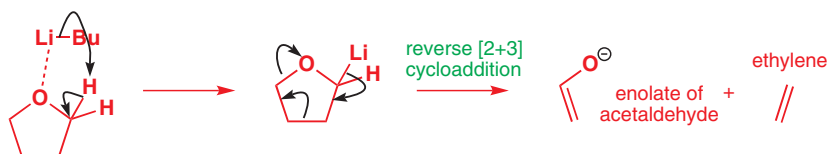
Ring-opening chemistry is characteristic of oxygen heterocycles too, and there is no need for us to revisit epoxide opening here. Epoxides are particularly reactive because ring-opening releases ring strain, driving the reaction forward. In general, though, oxygen heterocycles, as cyclic ethers, are relatively unreactive: ethers are the least reactive of all the common functional groups. This is one of the main reasons why THF and dioxane are such important solvents. A second reason is that they solvate organometallics by donating a lone pair to stabilize an electron-deficient metal cation (Li, for example). Cyclic ethers are better donors (more nucleophilic) than acyclic ones for the same reason that cyclic amines are more nucleophilic than acyclic ones.

This interaction of the lone pair with a Lewis acid can be exploited to make ethers more reactive. BF_3 is commonly used to activate cyclic ethers towards nucleophilic attack, and even with epoxides it increases the rate and yield of the reaction when organometallic reagents are used as nucleophiles. $BuLi$ does not react with oxetane unless a Lewis acid, such as BF_3 , is added, when it opens the four-membered ring to give a quantitative yield of n -heptanol. Without ring strain to help the reaction along, THF, by contrast, gives only a low yield of the product even with a Lewis acid.



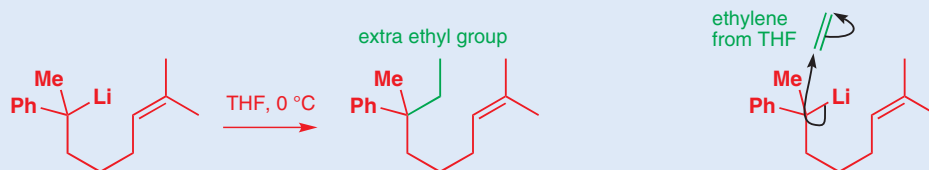
■ BF_3 is most easily handled as its complex with diethyl ether, written $BF_3 \cdot OEt_2$ or $BF_3 \bullet OEt_2$, in which the ether lone pair donates into the boron's empty p orbital. In related reactions HBr , BBr_3 , or Me_3SiCl are used to activate methyl and benzyl ethers of phenols towards nucleophilic attack. See Chapters 15, p. 351 and 23, p. 551.

A more common (if often unwanted) reaction between BuLi and THF is not nucleophilic attack, but deprotonation. You will have noticed that reactions involving BuLi in THF are invariably carried out at temperatures of 0°C or below—usually –78°C. This is because, at temperatures above 0°C, deprotonation of THF begins to take place. The deprotonated THF is unstable, and it undergoes a reaction we call a reverse [2 + 3] cycloaddition (see Chapter 34). Here is the mechanism (we have represented the organolithium as an anion to help with the arrows). The products are: (1) the (much less basic) enolate of acetaldehyde and (2) ethylene. The first tends to polymerize, and the second usually (but see the box below!) evaporates from the reaction mixture.



The case of the unexpected ethyl group

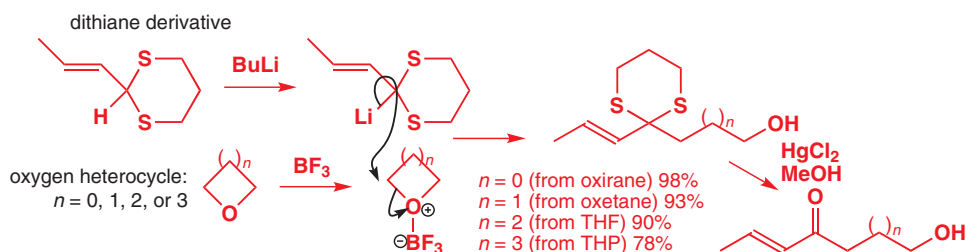
Some chemists in Belgium were studying the reactions of the organolithium shown here to find out whether the anionic centre would attack the double bond to form a five-membered ring. The reaction was slow, and they stirred the organolithium in THF for 6 hours at 0°C. When they worked the reaction up they found no five-membered ring products: instead they got a compound with an extra ethyl group! They showed that this ethyl group, in fact, comes from THF: the organolithium did not add to the double bond in the same molecule, but it did add slowly and in low yield to the double bond of the ethylene that is formed by decomposition of THF.



The most common use of tetrahydropyran derivatives is as protecting groups: you met this in Chapter 23.

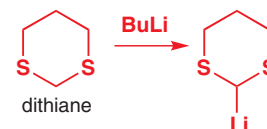
Sulfur heterocycles

As you saw in Chapter 27, sulfur stabilizes an adjacent anion, meaning that sulfur heterocycles are much easier to deprotonate than THF. The most important of these contains two sulfur atoms: dithiane. Deprotonation of dithiane occurs in between the two heteroatoms, and you saw some chemistry that arises from this on p. 661. The series of reactions below illustrates nicely both dithiane chemistry and the ring opening of oxygen heterocycles in the presence of BF₃. This substituted derivative of dithiane is deprotonated by BuLi to give a nucleophilic organolithium that will attack electrophiles—even oxygen heterocycles—provided BF₃ is present. The products are formed in excellent yield, even when the electrophile is THP, with no ring strain to drive the reaction. After the addition reaction the dithiane ring can be hydrolysed with mercury(II) to give a ketone carrying other useful functional groups.



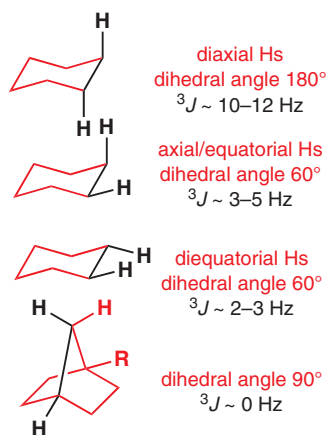
THF stability

The half-life of *n*-BuLi in THF (in the presence of TMEDA) is 40 minutes at 20°C, 5.5 hours at 0°C, and 2 days at –20°C. Diethyl ether is much less readily deprotonated: at 20°C in ether *n*-BuLi has a half life of 10 hours. With more basic organolithiums, the rate of decomposition of THF is even faster, and *t*-BuLi can be used in THF only at –78°C. At –20°C *t*-BuLi has a half-life in THF of only 45 minutes; in ether its half-life at this temperature is 7.5 hours.



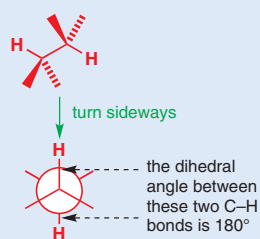
■ Dithiolane, the five-membered version of dithiane, cannot be used in this reaction because, although it is easy to deprotonate, the anion which forms decomposes by the same mechanism as lithiated THF.

➔ Coupling in alkenes is described on p. 293; coupling in cyclohexanes on p. 415.

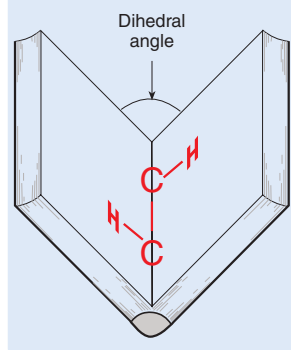


Dihedral angles

The dihedral angle is obvious in a Newman projection—it is the angle between the two C–H bonds projected on a plane orthogonal to the C–C bond. In a Newman projection this plane is the plane of the paper, and here the angle is 180° .



Another way to think of the dihedral angle is by imagining the C–C bond lying along the spine of a partially opened book. If the C–H bonds are written one on one page and the other on the other, then the dihedral angle is the angle between the pages of the book.

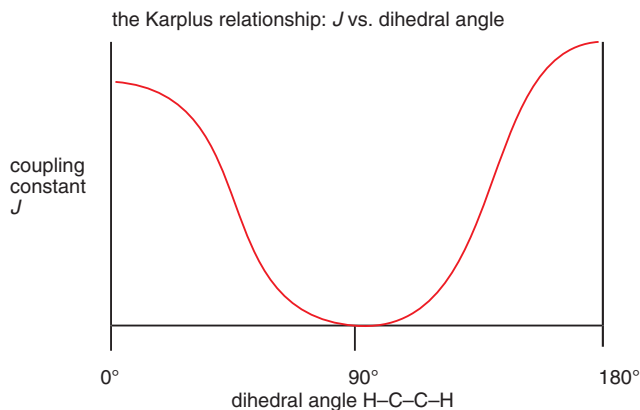


Conformation of saturated heterocycles

Using NMR to study conformation: the Karplus relationship

In Chapters 13 and 18 we explained that coupling in NMR spectra is a *through-bond* (and not a *through-space*) effect—that is why *trans* alkenes have bigger coupling constants than *cis* alkenes, and why axial–axial coupling in six-membered rings is larger than axial–equatorial or equatorial–equatorial coupling. We now need to build some more detail into your understanding of the relationship between conformation and coupling constants so we can use NMR to probe the conformations adopted by saturated rings.

The coupling constants in a cyclohexane tell us that coupling is greatest when the C–H bonds involved are most parallel—in other words when their dihedral angle is close to 180° or 0° . C–H bonds in simple cyclohexanes can have dihedral angles of only 60° or 180° , but by examining coupling constants in a range of other compounds, it is possible to draw up a description of the way coupling varies with dihedral angle. For example, in the bicyclic compound in the margin, the black protons have a dihedral angle close to 90° and the coupling constant is 0 Hz. The complete correlation was worked out by Karplus in the 1960s and is called the Karplus relationship. It is easiest to understand as a graph of J against dihedral angle.



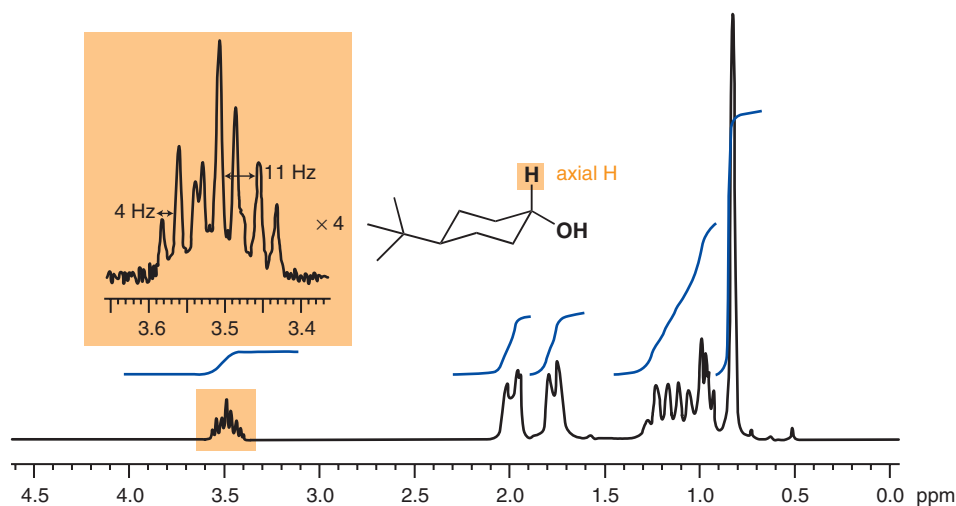
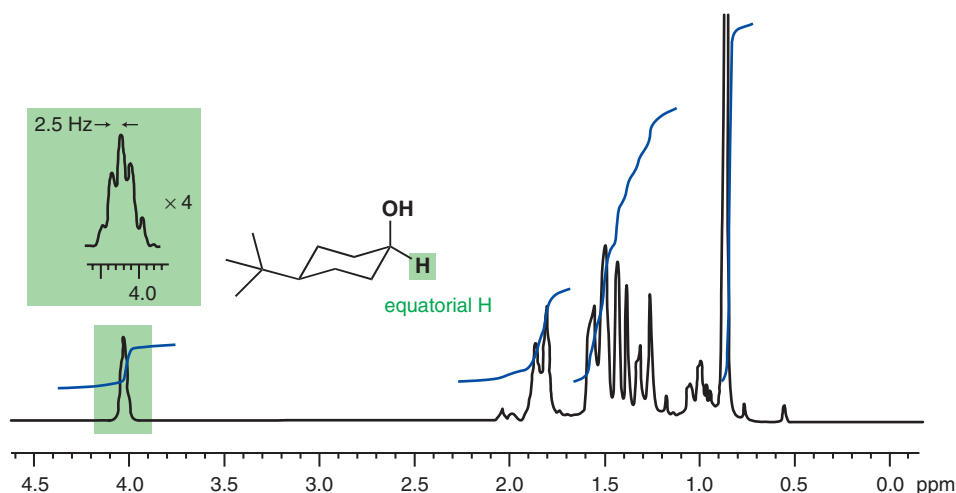
Examine the graph above carefully and note these principal features:

- Coupling is largest at 180° when the orbitals of the two C–H bonds are perfectly parallel (the situation in a *trans* alkene or the *trans*-diaxial C–H bonds of a cyclohexane).
- Coupling is nearly as large at 0° when the orbitals are in the same plane but not parallel (the situation in a *cis* alkene).
- Coupling is zero when the dihedral angle is 90° —orthogonal orbitals do not interact.
- The curve is flattened around 0° , 90° , and 180° — J varies little in these regions from compound to compound.
- The curve slopes steeply at about 60° and 120° — J varies a lot in this region with small changes of angle and from compound to compound.
- Numerical values of J vary with substitution, ring size, etc., but the Karplus relationship still works—it gives good *relative* values.

The determination of *conformation* by NMR may determine *configuration* at the same time. This often occurs when there are two or more substituents on the ring. Here is a simple example: you saw in Chapter 16 that the reduction of 4-*t*-butylcyclohexanone can be controlled by choice of reagent to give either a *cis* or a *trans* alcohol.



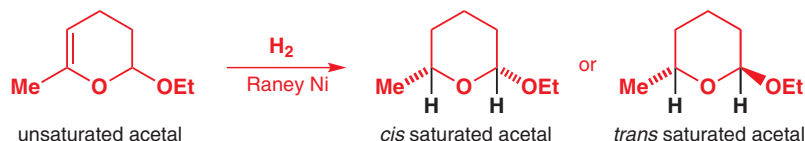
The products are easy to tell apart because the green H appears quite different in the NMR spectrum in the two cases. In one it is quite a fine multiplet; in the other it is much broader.



The bulky *t*-butyl group always goes equatorial, and each OH group has two identical axial neighbours and two identical equatorial neighbours (two are shown in black in the scheme at the bottom of p. 796—there are two more at the front). Each coloured H appears as a triplet of triplets. In the *cis* alcohol both couplings are small (2.72 and 3.00 Hz) but in the *trans* alcohol the axial–axial coupling is much larger (11.1 Hz) than the axial–equatorial (4.3 Hz) coupling.

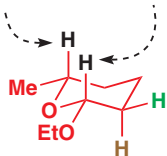
The same ideas can be used to study conformation in saturated *heterocyclic* systems. Hydrogenation of the double bond in this unsaturated acetal gives the saturated compound as a single isomer. But which one? Are the two substituents, Me and OEt, *cis* or *trans*?

You can draw a general conclusion from this observation: an NMR signal is roughly as wide as the sum of all its couplings. In any given compound, an axial proton will have a much *wider* signal than an equatorial proton.



The appearance of the two black hydrogens in the NMR spectrum reveals the answer and also shows what conformation the molecule adopts. There is a 1H signal at 3.95 ppm (which is therefore next to oxygen) and it is a double quartet. It must be the hydrogen next to the methyl group because of the quartet coupling. The quartet coupling constant has the 'normal'

δ_{H} 3.95, 1H, dq, J 9 and 6.5 Hz
 δ_{H} 4.40, 1H, dd, J 9 and 2 Hz

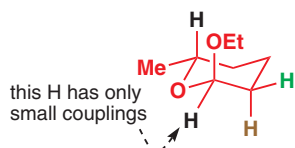


J value of 6.5 Hz. The doublet coupling is 9 Hz and this is too large to be anything other than an axial–axial coupling. This hydrogen is axial.

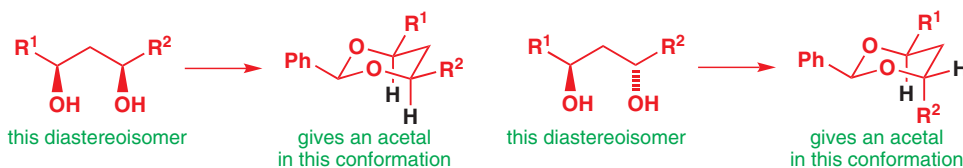
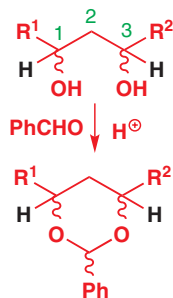
There is another 1H signal at 4.40 ppm (next to *two* oxygens), which is a double doublet with $J = 9$ and 2 Hz. This must also be an axial proton as it shows an axial–axial (9 Hz) and an axial–equatorial coupling. We now know the conformation of the molecule.

Both black hydrogens are axial so both substituents are equatorial. That also means in this case that they are *cis*. But note that this is because they are both on the same, upper side of the ring, not because they are both equatorial! The hydrogen at the front has two neighbours—an axial (brown) H, $J = 9$, and an equatorial (green) H, $J = 2$ Hz. All this fits the Karplus relationship as expected. You may have spotted that the H at the back appears to be missing a small coupling to its equatorial neighbour. No doubt it does couple, but that small coupling is not noticed in the eight lines of the double quartet. Small couplings can easily be overlooked.

When this compound is allowed to stand in slightly acidic ethanol it turns into an isomer. This is the *trans* compound and its NMR spectrum is again very helpful. The proton next to the methyl group is more or less the same but the proton in between the two oxygen atoms is quite different. It is at 5.29 ppm and is an unresolved signal of width about 5 Hz. In other words it has no large couplings and must be an equatorial proton. The conformation of the *trans* compound is shown in the margin.



■ This switch of the OEt group from the (usually favoured) equatorial to the axial position may seem odd, but will be explained in the next section.

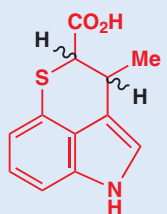


Now the molecule has a fixed conformation and the coupling constants of the black Hs to the neighbouring CH_2 group can be determined—an axial H will show one large J value, an equatorial H only small J values.

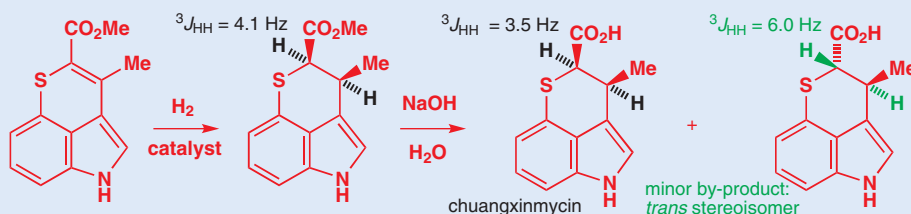
Deducing the stereochemistry of a new antibiotic

Only fully saturated six-membered rings are really chairs or boats. Even with one double bond in the ring, the ring is partly flattened: here we will look at an even flatter example. A unique antibiotic has been discovered in China and called 'chuangxinmycin' (meaning 'a new kind of mycin' where mycin = antibiotic). It is unique because it is a sulfur-containing indole: few natural products and no other antibiotics have this sort of structure.

The structure itself was easy to elucidate, but the stereochemistry of the two black hydrogens was not so obvious. The coupling constant (3J) was 3.5 Hz. During attempts to synthesize the compound, Kozikowski hydrogenated the alkene ester below to give an undoubted *cis* product (hydrogenation is *cis* selective: see Chapter 23, p. 535).



chuangxinmycin

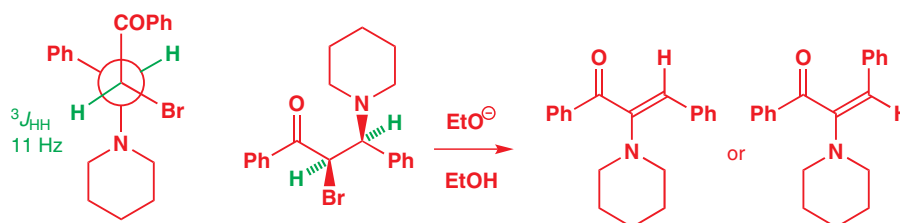


The 3J coupling between the black hydrogens in this compound was 4.1 Hz, much the same as in the antibiotic and, when the ester group was hydrolysed in aqueous base, the main product was identical to natural chuangxinmycin. However, there was a minor product, which was the *trans* isomer. It had $^3J = 6.0$ Hz. Note how much smaller this value is than the axial–axial couplings of 10 Hz or more in saturated six-membered rings. The flattening of the ring reduces the dihedral angle, reducing the size of J .

Coupling constants do not always give unambiguous information about stereochemistry, and in the next section we look at one technique which allows structural information to be extracted from NMR spectra without relying on coupling.

Determining stereochemistry when coupling constants are no help: the nuclear Overhauser effect

The coupling constant between the green protons of the compound below is rather large, at 11 Hz—about the same as the *trans* diaxial coupling in a cyclohexane. The Karplus relationship suggests the green protons must therefore spend much of their time with their bonds arranged with a dihedral angle close to 180° , and from this we can deduce that the compound has the conformation, as well as the configuration, shown. A more difficult problem is the assignment of the stereochemistry of the elimination product from this bromoamine and base. It's not a simple question, because the elimination also involves rearrangement of the amino group. The product is an alkene with two possible geometries.



Usually we would use coupling constants to determine alkene geometry, but they are no use here as there is only one proton on the alkene: it will be a singlet in both compounds. In such cases, we can make use of a quirk of NMR known as the **nuclear Overhauser effect** (NOE). NOE is rather different from coupling in the information it provides: it tells us which hydrogens are **close in space** rather than their relationship through bonds as revealed by coupling constants.

The details of the origin of the nuclear Overhauser effect are beyond the scope of this book, but we can give you a general idea of what the effect is. As you learned in Chapters 3 and 13, when a proton NMR spectrum is acquired, a pulse of radio frequency electromagnetic radiation jolts the spins of the protons in the molecule into a higher energy state. The signal we observe is generated by those spins dropping back to their original states. So far we have assumed that the drop back down is spontaneous, just like a rock falling off a cliff. In fact it isn't—something needs to 'help' the protons to drop back again—a process called **relaxation**. And that 'something' is other nearby magnetically active nuclei—usually more protons. Notice *nearby*—nearby in space not through bonds. With protons, relaxation is always fast, and the number of nearby protons does not affect the appearance of the NMR spectrum.

Although in a normal spectrum peak intensity is independent of the number of nearby protons, by using methods whose description is beyond the scope of this book it is possible to modify the intensity of the peaks very slightly according to the number of protons that are nearby. The basis of the method is that certain protons (or groups of identical protons) are irradiated selectively (in other words, they are jolted into their high energy state and held there by a pulse of radiation at exactly the right frequency—not the broad pulse needed in a normal NMR experiment). Under the conditions of the experiment, this causes protons that *were* relying on those irradiated protons to relax them to appear as a slightly more intense (by maybe just a few per cent) peak in the NMR spectrum. This effect is known as the nuclear Overhauser effect, and the increase in intensity of the peak the nuclear Overhauser enhancement. Both are shortened to 'NOE'.

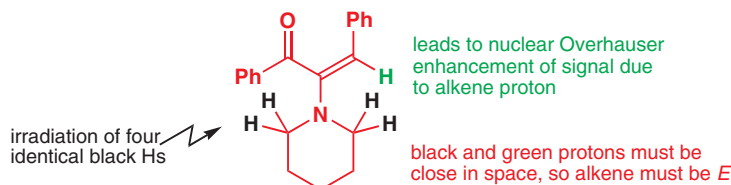
➔ See p. 293 for details of how to determine alkene geometry using the size of coupling constants.

Why you can't integrate ^{13}C NMR spectra

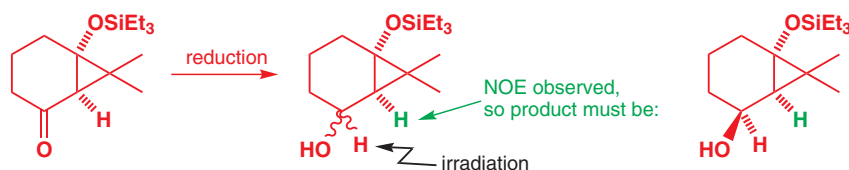
Relaxation is the real reason why you can't integrate ^{13}C signals. Relaxation of ^{13}C is slow, but is fastest with lots of nearby protons. This is the reason that you will often find that $-\text{CH}_3$ groups show strong signals in the ^{13}C NMR, while quaternary carbons, with no attached protons, show weak ones: quaternary carbons relax only slowly, so we don't detect such an intense peak. Allowing plenty of time for all ^{13}C atoms to relax between pulses gives more proportionally sized peaks, but at the expense of a very long NMR acquisition time.

All you need to be aware of at this stage is that irradiating protons in an NOE experiment gives rise to enhancements at other protons that are nearby in space—no coupling is required, and NOE is *not* a through-bond phenomenon. The effect also drops off very rapidly: the degree of enhancement is proportional to $1/r^6$ (where r is the distance between the protons) so moving two protons twice as far apart decreases the enhancement one can give to the other by a factor of 64. NOE spectra are usually presented as differences: the enhanced spectrum minus the unenhanced, so that the small enhancements of intensity in the peaks of certain protons can be spotted immediately.

Applying NOE to the problem in hand solves the structure. If the protons next to the nitrogen atom in the piperidine ring are irradiated, the signal for the alkene proton increases in intensity, so these two groups of protons must be near in space. The compound is the *E* alkene.

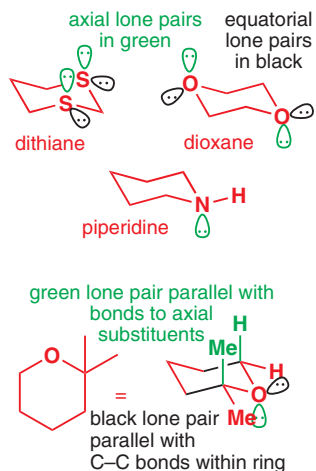


Data from NOE experiments nicely supplement information from coupling constants in the determination of three-dimensional stereochemistry too. Reduction of this bicyclic ketone with a bulky hydride reducing agent gives one diastereoisomer of the alcohol, but which? Irradiation of the proton next to the OH group leads to an NOE to the green proton. This suggests that the two protons are on the same side of the molecule and that reduction has occurred by hydride delivery to the face of the ketone opposite the two methyl groups on the three-membered ring.



A combination of coupling constants and NOE effects is routinely used to assign the stereochemistry of reaction products.

Heteroatoms in rings have axial and equatorial lone pairs



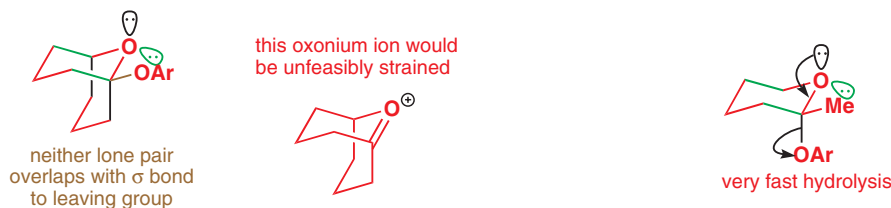
To a first approximation, the conformation of five- and six-membered saturated heterocycles follows very much the same principles as the conformation of carbocyclic compounds that we detailed in Chapter 16. For dithiane the conformation is as shown in the margin. Since the sulfur atoms have lone pairs, they too occupy axial and equatorial positions. The same is true of dioxane or of piperidine.

We have coloured the lone pairs green or black according to whether they are axial or equatorial, but you can also consider the colour coding in a different way: black lone pairs are parallel with C—C or C—heteroatom bonds in the ring; green lone pairs are parallel with axial C—H bonds outside the ring, or, if the ring has substituents, with the bonds to those substituents. This substituted tetrahydropyran illustrates all this. Notice that the equatorial substituents next to the heteroatom are parallel with neither the green nor the black lone pair.

Why is this important? There are many reactions in which lone pairs have an important role to play. For example, in an acetal hydrolysis, stabilization of the forming positive charge by an adjacent lone pair facilitates the elimination step of the mechanism. Let's consider what happens in this acetal hydrolysis where the acetal is a saturated heterocycle. From Chapter 11, you expect this to be the mechanism:



Yet when we try to draw the conformation of the lone pairs we run into a problem: neither overlaps with the C–O bond that is breaking and so neither can donate its electron density into the C–O σ^* . Another way of looking at this is to say that the intermediate oxonium ion—with a C=O double bond formed by one of the oxygen's lone pairs—would be extremely strained. Not surprisingly, the rate of hydrolysis of this acetal is very slow compared with similar ones in which overlap between the oxygen lone pair and the C–O σ^* is possible. The acetal on the right hydrolyses about 10^{10} times faster.



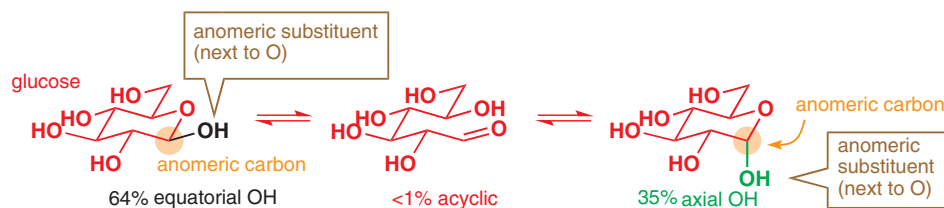
You have just seen that overlap between orbitals governs NMR coupling constants; other situations where orbital overlap is important are:

- E2 elimination reactions (Chapter 17)
- reactions of cyclic molecules (Chapter 32)
- the Felkin–Anh transition state conformation (Chapter 33)
- fragmentations and rearrangements (Chapter 36).

Together, these effects are called *stereoelectronic effects* because they all depend on the *orientation* of orbitals.

Some substituents of saturated heterocycles prefer to be axial: the anomeric effect

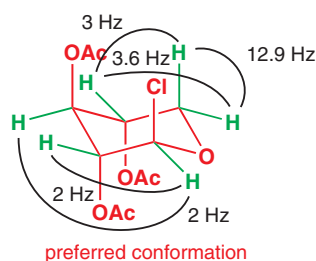
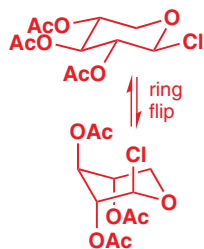
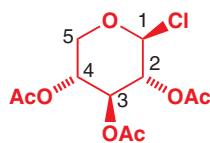
Many of the stereoelectronic effects in the list above govern *reactivity*, but the next section will deal with how stereoelectronic effects affect *structure*—and in particular *conformation*. Some of the most important saturated oxygen heterocycles are the sugars. Glucose is a cyclic hemiacetal—a pentasubstituted tetrahydropyran if you like—whose major conformation in solution is shown below. About two-thirds of glucose in solution exists as this stereoisomer, but hemiacetal formation and cleavage is rapid, and this is in equilibrium with a further one-third that carries the hemiacetal hydroxyl group axial (<1% is in the open-chain form).



Having read Chapter 16 you will not be surprised that glucose prefers all its substituents to be equatorial. For four of them, of course, there is no choice: they are either all-equatorial or all-axial, and the only way they can get from one to the other is by ring-flipping. But for the fifth substituent, the hydroxyl group next to the ring oxygen (known as the anomeric hydroxyl group), a choice between axial or equatorial is made available by hemiacetal cleavage and re-formation—it can invert its configuration. What is perhaps surprising is that the equatorial preference of this hydroxyl group is so small—only 2:1. Even more surprising is that, for most derivatives of glucose, the anomeric substituents *prefer* to be axial rather than equatorial.



➡ We introduced the hemiacetal structure of glucose in Chapter 6, p. 137.



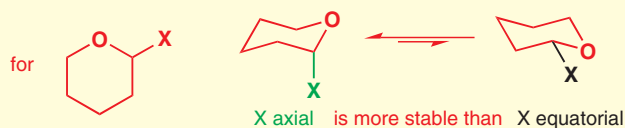
Move away from glucose and the effect is still there in other substituted tetrahydropyrans. Listed below are the NMR signals of the chloro compound in the margin. There are now only two possible conformations (no configurational changes are possible because this is not a hemiacetal)—both shown—and from the NMR spectrum you should be able to work out which one this compound has.

δ			J , Hz	
5.78	1H	t	2.0	H1
5.03	2H	m		H2, H3
4.86	1H	m		H4
4.37	1H	dd	12.9, 3.0	H5a
3.75	1H	ddd	12.9, 3.7, 0.6	H5b
2.10	9H	s		OAc $\times$ 3

The key point is that axial–axial couplings are large (>8 Hz, say), even with adjacent electronegative atoms (which tend to lower coupling constants). So if H1 were an axial proton, you would expect it to have a large coupling to H2. But it doesn't—it couples to H2 with J of only 2.0 Hz. (The other coupling is a W-coupling to H3, also of 2.0 Hz: see p. 296.) Similarly, we know that the 12.9 Hz coupling shared by the two H5 protons must be a geminal (2J) coupling. One of H5a or H5b must be axial, yet both couple to H4 with $J < 4$ Hz, so H4 cannot be axial. With this evidence, we have to conclude that H1 and H4 (and therefore H2 and H3) are equatorial, so the compound must exist mainly in the all-axial conformation. (The 0.6 Hz coupling to H5b is another W-coupling, and shows that H5b is the equatorial proton and H5a therefore the axial one.) This axial preference is called the **anomeric effect**.

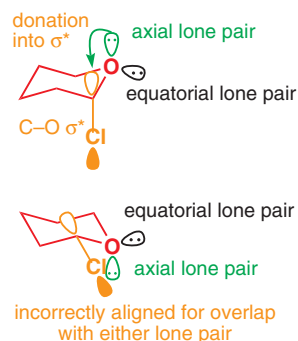
• The anomeric effect

In general, any tetrahydropyran bearing an electronegative substituent in the 2-position will prefer that substituent to be axial. This is known as the **anomeric effect**.



But why? This goes against all of what we said in Chapter 16 about axial substituents being more hindered, making conformations carrying axial substituents disfavoured. The key again is stereoelectronics, and we can now link up with the message we left you with at the end of the last section: elimination reactions are possible only when the orbitals involved are parallel.

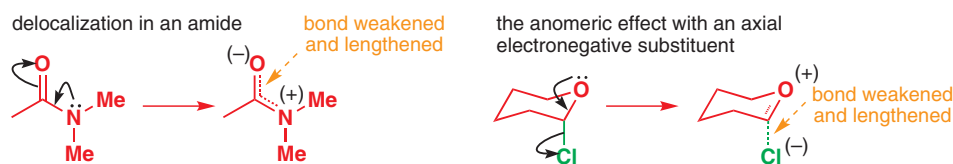
An amide is more stable (less reactive) than a ketone because the p orbital of the N and the low-lying $C=O$ π^* of the carbonyl can lie parallel—they can overlap and electron density can move from nitrogen into the $C=O$ bond, weakening $C=O$. (Evidence for this comes from the lower IR stretching frequency of an amide $C=O$, among other things.) But $C-X$ bonds also have low-lying antibonding orbitals—the $C-X$ σ^* —so we would expect a molecule likewise to be stabilized if an adjacent heteroatom could donate electrons into this orbital. Take the generalized tetrahydropyran in the box above, for example, with $X=Cl$, say. This molecule is most stable if an oxygen lone pair can overlap with $C-Cl$ σ^* , as shown in the margin.



But it can do this only if the chlorine is axial! Remember what we pointed out earlier: the oxygen's equatorial lone pairs are parallel with nothing but bonds in the ring, so the oxygen's axial lone pair is the only one that can help stabilize the molecule, and it can only do this when the Cl is axial. Only the axial conformation benefits from the stabilization, and this is the origin of the anomeric effect.

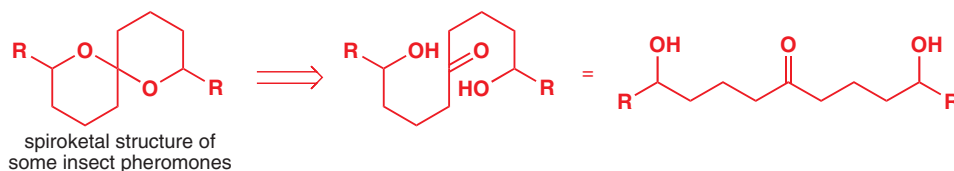
How shall we represent the stabilization? Comparing again with the amide stabilization, you might think about how to represent it with curly arrows: this is straightforward with the amide and you have seen it many times. But it looks odd with our heterocycle: electron density moves

from O to Cl, and the C–Cl bond is weakened. If the process carried right on, Cl^- would leave. This is exactly what did happen in the acetal we presented you with as an example on p. 801: only the axial OAr could leave, however, because of the same requirement for overlap with an oxygen lone pair. In the real structure that we are now looking at, the Cl is still there: the C–Cl bond is weaker, and some of the oxygen's electron density is delocalized on to Cl. This can be seen in crystal structures: compounds exhibiting an anomeric effect have a longer (and therefore weakened) bond outside the ring and a shorter, stronger C–O bond within the ring.



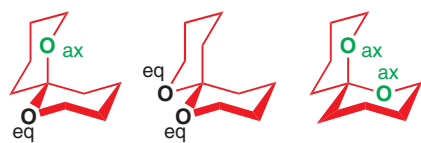
The anomeric effect in spiroketals

Now that you know about the anomeric effect, you should add it to your mental array of possible ways to explain 'unexpected' results. Here is an example. Many fruit flies have pheromones based around a 'spiroketal' structure, which we could represent without stereochemistry as shown below. You can imagine the spiroketal (that is, an acetal of a ketone made of two rings joined at a single atom) being made from a dihydroxyketone—and, indeed, this is very often how they are made synthetically. But this is a bad representation because these compounds do have stereochemistry, and the stereochemistry is very interesting.

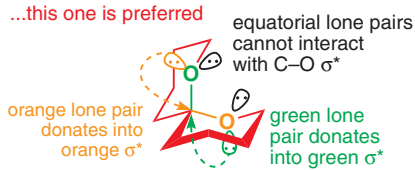


Let's start with the simplest example, with $\text{R}=\text{H}$ (a pheromone of the olive fly). Once you have drawn one ring in its chair conformation, there are three ways of attaching the other ring, shown here. If you think they all look the same, consider the orientation of each C–O bond with respect to the ring that it is not part of: you can have each C–O axial or equatorial, and there are three possible arrangements (three conformations).

of the three conformations of the spiroketal...



...this one is preferred



Without knowing about the anomeric effect, you would find it hard to predict which conformation is favoured, and, indeed, you might expect to get a mixture of all three. But NMR tells us that this compound exists entirely in one conformation: the last one here, in which each oxygen is axial on the other ring. Only in this conformation can both C–O bonds benefit from an anomeric effect—this is often known as the **double anomeric effect**.

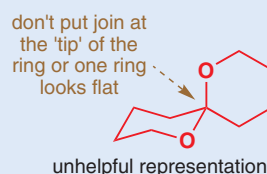
Related effects in other types of compounds

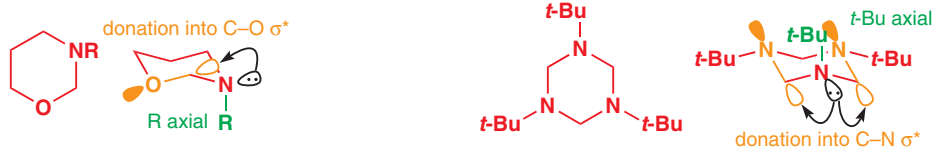
The key requirement for the anomeric effect is that there is a heteroatom with a lone pair (O, N, S usually) adjacent to (that is, in a position to interact with) a low-lying antibonding orbital—usually a C–X sigma* (where X=halogen or O). The C–X bond doesn't have to be within the ring—for example, the nitrogen heterocycle on the left prefers to have the R group axial so that the nitrogen gets an equatorial lone pair. Equatorial lone pairs are parallel with bonds within the ring, one of which is C–O, and this conformation is therefore stabilized by an N lone pair/C–O sigma* interaction.

■ This is a chiral compound, even though the acetal centre is not a chiral centre: no conformation has a plane of symmetry.

Hint on drawing spiroacetals

If you try to draw these spirocyclic acetals you will soon find there is a trick to getting them to look right: the *spiro* carbon has to be one of the four that aren't at the 'point' of either ring; otherwise one ring ends up looking flat.

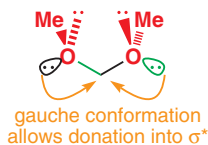
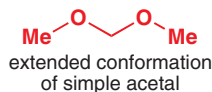
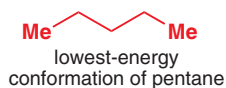
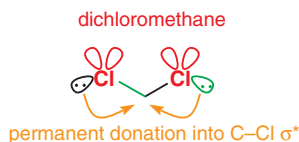




It would be a bit much for the 1,3,5-triazine on the right to have all three *t*-butyl groups axial (too much steric hindrance), but it can get away with having one of them axial, benefitting from the resulting equatorial lone pair, which can overlap with two C–N σ^* s in the ring.

It's not only in six-membered rings that stereoelectronic interactions between filled and unfilled orbitals stabilize some conformations more than others. Stereoelectronic effects control the conformations of many types of molecules.

● Any conformation in which a lone pair is *anti*-periplanar to a low-energy antibonding orbital will be stabilized by a stereoelectronic interaction.



➡ Terms used to describe conformations (gauche, synclinal etc.) are defined on p. 365.

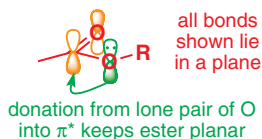
We shall look at three common compounds that are stabilized by stereoelectronic effects: in two cases, the stabilization is specific to one conformation, and we can use stereoelectronics to explain what would otherwise be an unexpected result.

We start with a compound that is so simple that it has only one conformation because it has no rotatable bonds: dichloromethane. You may have wondered why it is that, while methyl chloride (chloromethane) is a reactive electrophile that takes part readily in substitution reactions, dichloromethane is so unreactive that it can be used as a solvent in which substitution reactions of other alkyl halides take place. You may think that this is a steric effect: indeed, Cl is bigger than H. But CH_2Cl_2 is much less reactive as an electrophile than ethyl chloride or propyl chloride: there must be more to its unreactivity. And there is: dichloromethane benefits from a sort of 'permanent anomeric effect'. One lone pair of each chlorine is always *anti*-periplanar to the other C–Cl bond so that there is always stabilization from this effect.

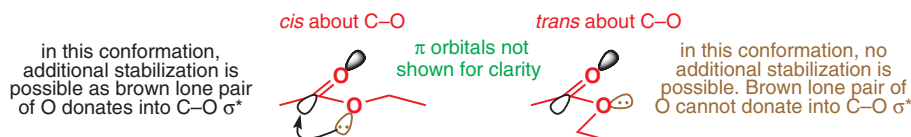
Among the most widespread classes of acyclic compounds to exhibit stereoelectronic control over conformation are acetals. Take the simple acetal of formaldehyde and methanol, for example: what is its conformation? An obvious suggestion is to draw it fully extended so that every group is fully *anti*-periplanar to every other—this would be the lowest energy conformation of pentane, which you get if you just replace the Os with CH_2 s.

The trouble is, in this conformation none of the oxygen lone pairs get the chance to donate into the C–O σ^* orbitals. Although putting the bonds *anti*-periplanar to one another makes steric sense, electronically, the molecule much prefers to put the lone pairs *anti*-periplanar to the C–O bonds, so the bonds themselves end up *gauche* (synclinal) to one another. This is known as the *gauche* effect, but is really just another way in which the stereoelectronic effects that give rise to the anomeric effect turn up in acyclic systems.

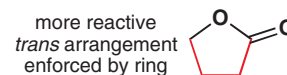
Finally, an even more familiar example that you may never have thought about. You are well aware now that amides are planar, with partially double C–N bonds, and that tertiary amides have one alkyl group *cis* to oxygen and one *trans*. But what about esters? Esters are less reactive than acyl chlorides because of donation from the oxygen p orbital into the carbonyl π^* , so we expect them to be planar too, and they are. But there are two possible planar conformations for an ester: one with R *cis* to oxygen and one with R *trans*. Which is preferred?



Here are the two conformations drawn out for ethyl acetate. When the ethyl group ($=R$) and O are *cis*, not only can one oxygen lone pair interact with the $C=O$ π^* , but the other lone pair can also donate into the σ^* of the $C=O$ bond. This is not possible when Et and O are *trans*: they are no longer anti-periplanar. The *cis* conformation of esters is generally the preferred one, even in formate esters, where the alkyl group ends up in what is clearly a more sterically hindered orientation.



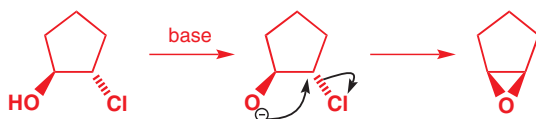
Cyclic esters—lactones—cannot lie *cis* because of the ring, and this is one of the reasons why lactones are distinctly more reactive than esters and in many reactions behave more like ketones: lactones are quite easy to reduce with $NaBH_4$, for example.



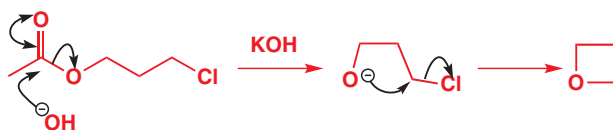
Making heterocycles: ring-closing reactions

We have talked about the structure of saturated heterocycles, particularly with regard to stereoelectronic control over conformation, and before that we looked at some of their reactions. We will now look at how to *make* them. By far the most important way of making them is by ring-closing reactions because we can usually use the heteroatom as the nucleophile in an intramolecular substitution or addition reaction. Ring-closing reactions are, of course, just the opposite of the ring-opening reactions we talked about earlier in the chapter, and we can start with a reaction that works well in both directions: ring closure to form an epoxide. You know well that epoxides can be formed using *m*-CPBA and an alkene, but you have already seen examples where they form by an intramolecular substitution reaction such as this.

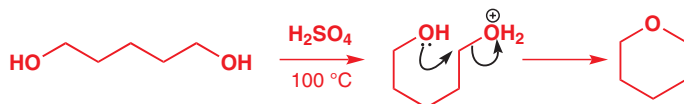
➔ *m*-CPBA epoxidation is discussed in Chapter 19, p. 429.



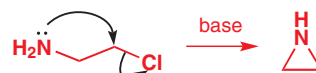
The same method can also be used to generate larger cyclic ethers. Oxetane, for example, is conveniently made by adding 3-chloropropyl acetate to hot potassium hydroxide. The first step in this reaction is the hydrolysis of the ester. The alkoxide produced then undergoes an intramolecular substitution reaction to yield oxetane.



Tetrahydropyran was prepared as early as 1890 by a ring closure that occurs when a mixture of 1,5-pentanediol with sulfuric acid is heated.



These are all S_N2 reactions, so you will not be surprised that nitrogen heterocycles can be prepared in the same way. Aziridine itself, for example, was first prepared in 1888 from 2-chloroethylamine. Related reactions can be used to form three-, five-, and six-membered nitrogen heterocycles, but fail to form four-membered rings. In fact, four-membered rings are generally among the hardest of all to form.



To illustrate this, the green columns of the table below show the rates (relative to six-membered ring formation = 1) at which bromoamines of various chain lengths cyclize to saturated nitrogen heterocycles of three to seven members.

Ring size	Product ^a	Relative rate ^a	Product ^b	Relative rate ^a	Assessment of rate
3		0.07			moderate
4		0.0001		0.58	slow
5		100		833	very fast
6		1		1	fast
7		0.002		0.0087	slow
8				0.00015	very slow

^aRelative to six-membered ring formation; ^bE = CO₂Et

At first sight it may seem that these rates have been produced by a random number generator! There seems to be no rhyme or reason to them, and no consistent trend. To convince you that these numbers mean something, the table also shows, in the orange columns, the relative rates for another ring-closing reaction, this time forming four- to seven-membered rings that are not even heterocycles by intramolecular alkylation of a substituted malonate. Although the numbers are quite different in the two cases, the ups and downs are the same, and the final column summarizes the relative rates. Put another way, a rough guide (only rough—it doesn't work in all cases) to the rate of ring formation is this.

● **Rough guide to the rate of formation of saturated rings**

Fastest 5 > 6 > 3 > 7 > 4 > 8–10 slowest

■ Remind yourself of our definition of small, normal, medium, and large rings, and what ring strain means, by re-reading p. 366. We will deal with what happens in large rings a little later.

We show the numbers in colour to highlight the fact that this seemingly illogical ordering of numbers actually conceals two superimposed trends. Once you get to five-membered rings, the rate of formation drops consistently as the ring size moves from 'normal' (5 and 6) to 'medium' (8 to 13) sized rings. 'Small' (3 and 4) rings insert into the sequence after 6.

The reason for the two superimposed trends is two opposing factors. Firstly, small rings form slowly because forming them introduces ring strain. This ring strain is there even at the transition state, raising its energy and slowing down the reaction. The activation energy for forming

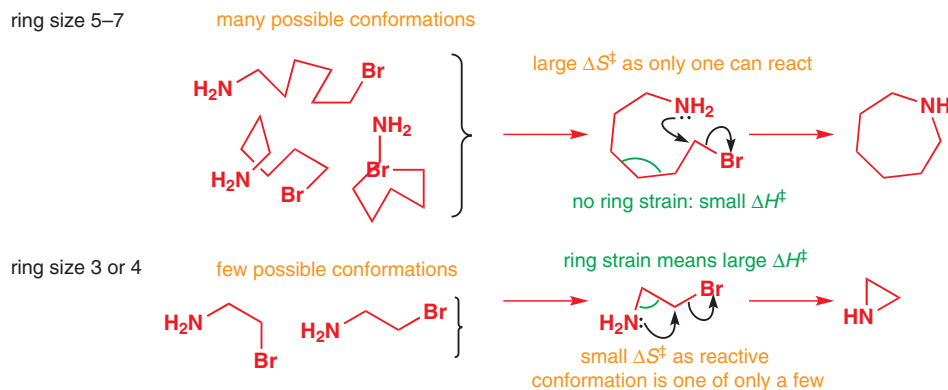
a three-membered ring is very high, due to strain, but decreases as the ring gets larger. This explains why three- and four-membered rings don't fit straightforwardly into the sequence.

But if the reaction rate simply depended on the strain of the product, the slowest reaction would be the formation of the three-membered ring, and six-membered rings (which are essentially strain-free) would form fastest. Yet the data shows that four-membered rings form more slowly than three-membered ones, and five-membered ones faster than six-membered ones. To explain this, we need to remind you of an equation we presented in Chapter 12.

$$\Delta G^\ddagger = \Delta H^\ddagger - T\Delta S^\ddagger$$

The activation energy barriers $\Delta G^\ddagger$ of our reactions are made up of two parts: an enthalpy of activation $\Delta H^\ddagger$, which tells us about the energy required to bring atoms together against the strain and repulsive forces they usually have, and an entropy of activation $\Delta S^\ddagger$, which tells us about how easy it is to form an ordered transition state from a wriggling and randomly rotating molecule.

$\Delta G^\ddagger$ for three- and four-membered ring formation is large because $\Delta H^\ddagger$ is large: energy is needed to bend the molecule into the strained small-ring conformation. $\Delta H^\ddagger$ for five-, six-, and seven-membered rings is smaller: this is the quantifiable representation of the 'ring strain' factor we have just introduced. The second factor is one that depends on $\Delta S^\ddagger$: how much order must be imposed on the molecule to get it to react. Think of it this way: a long chain has a lot of disorder, and to get its ends to meet up and react means it has to give up a lot of freedom. So, for the formation of medium and large rings, $\Delta S^\ddagger$ is large and negative, contributing to a large $\Delta G^\ddagger$ and slow reactions. For three-membered rings, on the other hand, the reacting atoms are already very close together and almost no order needs to be imposed on the molecule to get it to cyclize: rotation about just one bond is all that is needed to ensure that the amine group is in the perfect position to attack the σ^* of the C–Br bond in our example above. $\Delta S^\ddagger$ is very small for three-membered rings so, while $\Delta H^\ddagger$ is large, there is little additional contribution from the $T\Delta S^\ddagger$ term and cyclization is relatively fast. Four-membered rings suffer the worst of both worlds: forming a four-membered ring introduces ring strain ($\Delta H^\ddagger$) and requires order ($\Delta S^\ddagger$) to be imposed on the molecule. They form very slowly as a result.



These results are summarized in the following box.

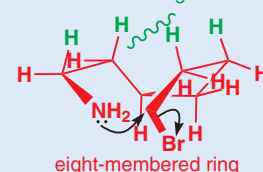
● Ring formation

- Three-membered ring formation is fast—the product is strained so $\Delta H^\ddagger$ is large but this is offset by the reacting atoms being as close as they can get in a freely rotating chain.
- Four-membered rings form slowly—the product is still significantly strained but the reacting atoms are now not right next to each other to offset this.
- Five-membered ring formation is often fastest of all. Significantly less strain and the ends are still not too far apart.
- Six-membered ring formation experiences no strain but neither does it have the advantage of the ends being close.
- Seven-membered rings and beyond form more slowly as $\Delta S^\ddagger$ increases.

Medium and large rings

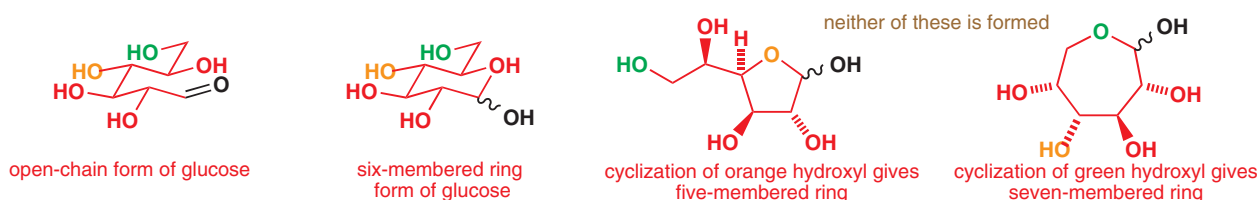
Beyond seven-membered rings, the rates of ring formation stay low but begin to level off, and may start to rise again when the rings have 10 or 11 members. These are the 'medium rings', of about 8–13 members, and they suffer from a different sort of strain, evident in the graph on p. 368 (Chapter 16), due to interactions between C–H bonds across the ring (transannular interactions). These are worst for rings of 8 and 9 members, and begin to be relieved once there are 10 or 11 atoms in the ring. For 14-membered rings and above, there is no transannular strain, and the rates of ring closure remain essentially constant at about the seven-membered ring mark. Rates of reactions in ring sizes of 14 and above are essentially little different from those in acyclic compounds. To get large rings to form, it is often necessary to carry out the cyclization reaction in very dilute solution to discourage competing intermolecular reactions.

transannular interactions hinder medium-ring formation

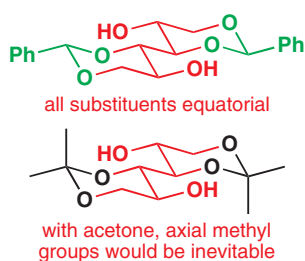
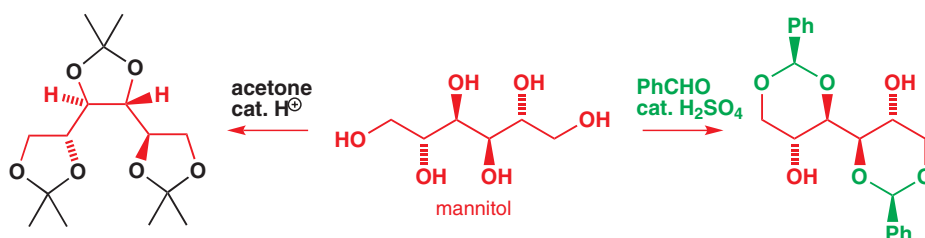


Thermodynamic control over ring size

In this section we have discussed the rate at which rings form: in other words the kinetics of ring formation. However, there are many ring-forming reactions that are under thermodynamic and not kinetic control. For example, you have already seen that glucose exists predominantly as a six-membered ring in solution. It could also exist as a five-membered ring: it doesn't because although five-membered rings form faster than six-membered ones, they are usually less stable (remember, a six-membered ring is essentially strain-free). For similar thermodynamic reasons, it doesn't exist as a seven-membered ring, even though you can draw a reasonable structure for it.



Thermodynamic control is important in other ways in carbohydrate chemistry because control over ring size allows selective protection of the hydroxyl groups of sugars. Compare these two reactions. Both of them give acetals from the same starting material, mannitol.



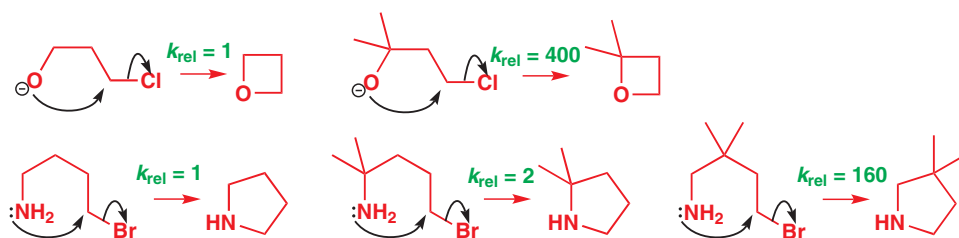
Don't be put off by the way in which we have had to twist half the molecule round to draw the left-hand structure: the stereochemistry hasn't changed. The important thing is that acetone reacts with mannitol to form three five-membered acetals (dioxolanes) while benzaldehyde forms only two six-membered acetals. This is quite a common result: when there is a choice, acetone prefers to react across a 1,2-diol to give a five-membered acetal, while aldehydes prefer to react across a 1,3-diol to form a six-membered acetal. Drawing a conformational diagram of the product on the right helps to explain why. All of the substituents are equatorial, making this a particularly stable structure. Now imagine what would happen if acetone formed this type of six-membered ring acetal. There would always be an axial methyl group, and the six-membered rings would be less stable.

Combatting $\Delta S^\ddagger$ —the Thorpe–Ingold effect

The rate of ring formation is affected not just by ring size but by substituents on the ring being formed. Compare the following relative rates (k_{rel}) for epoxide-forming cyclization reactions. The second looks as though it suffers more steric hindrance but nonetheless it is tens of thousands of times faster!



Adding substituents to other ring-forming reactions makes them go faster too: in the next two examples the products are oxetanes and pyrrolidines.

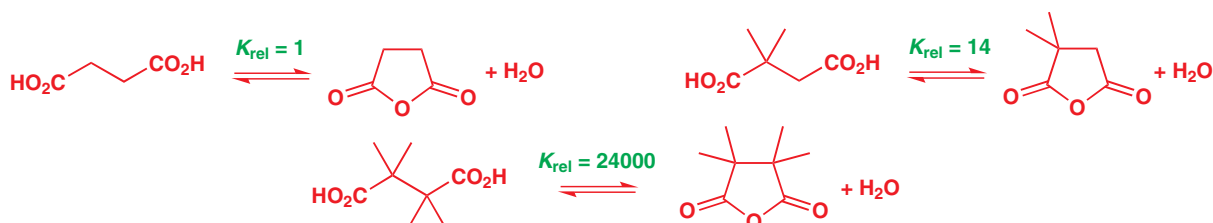


This effect is quite general, and is known as the Thorpe–Ingold effect after the first chemists to note its existence, in 1915.

● The Thorpe–Ingold effect

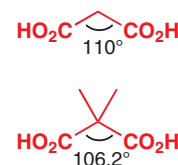
The Thorpe–Ingold effect is the way in which substituents on the ring increase the rate, or equilibrium constant, for ring-forming reactions.

As the box says, it's not only rate that can be affected by additional substitution. Here are the relative equilibrium constants for the formation of an anhydride from a 1,4-dicarboxylic acid (the unsubstituted acid is called succinic acid, and the values are scaled so that K_{rel} for the formation of succinic anhydride is 1). More substituents mean more cyclized product at equilibrium. The Thorpe–Ingold effect is both a kinetic and a thermodynamic phenomenon.



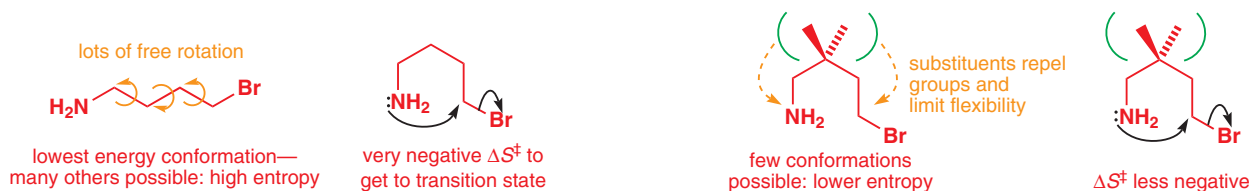
Now we need to explain why this is. The explanation comes in two parts, one of which may be more important than the other, depending on the ring being formed. The first part is more applicable to the formation of small rings, such as the first example we gave you.

If you measure the bond angles of chains of carbon atoms, you expect them to be close to the tetrahedral angle, 109.5° . The crystal structure of the 1,3-dicarboxylic acid in the margin, for example, shows a C–C–C bond angle of 110° . Now imagine adding substituents to the chain. They will repel the carbon atoms already there, and force them a little closer than they were, making the bond angle slightly less. X-ray crystallography tells us that adding two methyl groups to our 1,3-dicarboxylic acid decreases the bond angle by about 4° .



We can assume that the same is true in the alcohol starting materials for the epoxide-forming reactions on p. 808 (we can't measure the angle directly because the compounds aren't crystalline). Now consider what happens when both of these alcohols form an epoxide. The bond angle has to become about 60° , which involves about 50° of strain for the first diacid, but only 46° for the second. By distorting the starting material, the methyl groups have made it slightly easier to form a ring.

This part of the argument works only for small rings. For larger rings, we need another explanation, and it involves entropy. We'll use the pyrrolidine-forming reaction as an example. We have explained the effect of $\Delta S^\ddagger$ (entropy of activation) on the rate of ring formation: as larger rings form they have to lose more entropy at the transition state, and this contributes to a less favourable $\Delta G^\ddagger$.

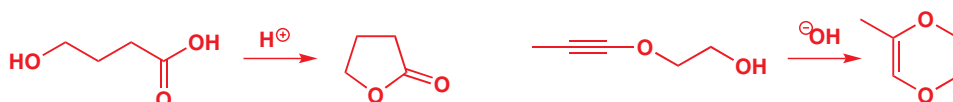


But, when the starting material has more substituents, it starts off with less entropy anyway. More substituents mean that some conformations are no longer accessible to the starting material—the green arcs on the structures on the right above show how the methyl groups hinder rotation of the N and CH_2Br substituents into that region of space. Of those fewer conformations, many approximate to the conformation in the transition state, and moving from starting material to transition state involves a smaller loss of entropy: $\Delta S^\ddagger$ is less negative so $\Delta G^\ddagger (= \Delta H^\ddagger - T\Delta S^\ddagger)$ is more negative and the ring forms faster. The same arguments apply to ΔS for the reaction as a whole (the difference in entropy between starting material and products), so increased substitution favours ring closure even under thermodynamic control.

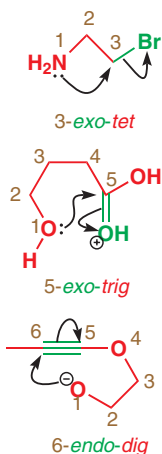
Baldwin's rules

Nearly all of the cyclization reactions that we have discussed have been intramolecular $\text{S}_\text{N}2$ reactions where one end of the molecule acted as the nucleophile displacing the leaving group on the other end. We kept to this sort of reaction in order to make valid comparisons between different ring sizes. But you can imagine making saturated heterocycles in plenty of other ways—intramolecular substitution at a carbonyl group, for example, such as happens in this lactonization reaction, or intramolecular addition of an oxyanion on to an alkyne.

➡ You saw a related alkyne addition in Chapter 27, p. 683.



Cyclization reactions can be classified by a simple system involving: (1) the ring size being formed, (2) whether the bond that breaks as the ring forms is inside (*endo*) or outside (*exo*) the new ring, and (3) whether the electrophile is an sp (digonal), sp^2 (trigonal), or sp^3 (tetrahedral) atom. This system places three of the cyclizations just shown in the following classes.



1. The ring being formed has three members; the breaking C–Br bond is outside the new ring (*exo*); the C carrying Br is a tetrahedral (sp^3) atom (*tet*).
2. The ring being formed has five members; the breaking C=O bond is outside the new ring (*exo*); the C being attacked is a trigonal (sp^2) atom (*trig*).
3. The ring being formed has six members; the breaking $\text{C}\equiv\text{C}$ bond is inside the new ring (*endo*); the C being attacked is a digonal (sp) atom (*dig*).

The classes of cyclization reactions are important, not because we have a compulsive Victorian desire to classify everything, but because which class a reaction falls into determines whether or not it is likely to work. Not all cyclizations are successful, even though they may look fine on paper! The guidelines that describe which reactions will work are known as *Baldwin's rules*: empirical observations backed up by some sound stereoelectronic reasoning. Reactions can be classified according to these rules as 'favoured' and 'disfavoured'. We will deal with the rules step by step and then summarize them in a table at the end.

Firstly, and not surprisingly (because we have been talking about them for much of this chapter):

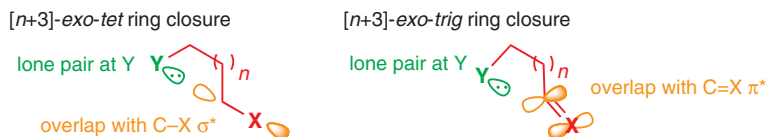
- all *exo-tet* cyclizations are favoured

and, similarly (again you can find many examples in this book):

- all *exo-trig* cyclizations are favoured

Interactive examples of Baldwin's rules

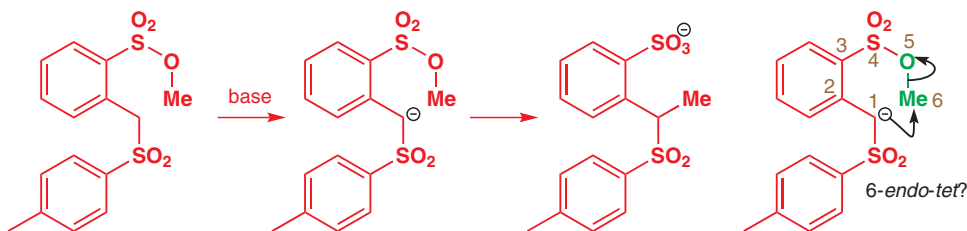
Despite the variation in rate we have described for this type of reaction, *exo-tet* cyclizations have no stereoelectronic problems: the lone pair and the C–X σ^* (X is the leaving group) can overlap successfully irrespective of ring size. The ring closures in the table on p. 809 all fall into this category. The same is true for *exo-trig* reactions: it is easy for the nucleophilic lone pair to overlap with the C=X π^* to form a new bond. Examples include lactone formation such as the one on p. 810.



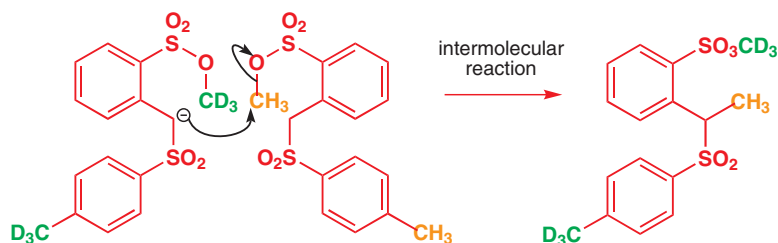
Endo-tet reactions are rather different. For a start:

● 5 and 6-*endo-tet* are disfavoured.

Endo-tet reactions would not actually make a ring, but they fall conveniently into the system and we will look at them here. Here is a reaction that looks as though it contradicts what we have just said. The arrows in the reasonable-looking mechanism on the right describe a 6-*endo-tet* process because the breaking Me–O bond is within the six-membered ring transition state (even if no ring is formed).



But Eschenmoser showed that, for all its appeal (intramolecular reactions usually outpace all alternatives), this mechanism is wrong. He mixed together the starting material for the reaction above with the hexadeuterated compound shown below, and re-ran the reaction. If the reaction had been intramolecular, the products would have contained either no deuterium, or six deuteriums. In the event, the product mixture contained about 25% of each of these compounds, with a further 50% containing three deuteriums. The products cannot have been formed intramolecularly, and this distribution is exactly what would be expected from an *intermolecular* reaction.



With *endo-trig* reactions, whether they work or not depends on the ring size.

● 3-, 4-, and 5-*endo-trig* are disfavoured; 6 and 7-*endo-trig* are favoured.

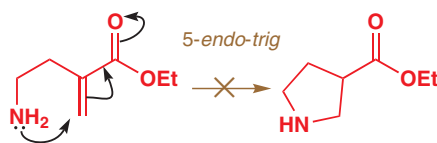
The most important case in the *endo-trig* class is the disfavoured 5-*endo-trig* reaction and, if there is one message you take away from this section, it should be that 5-*endo-trig* reactions are disfavoured. The reason we say this is that 5-*endo-trig* cyclizations are reactions that look perfectly fine on paper, and at first sight it seems quite surprising that they won't work. This intramolecular conjugate addition, for example, appears to be a reasonable way of making a substituted pyrrolidine.

Professor Sir Jack Baldwin worked in Oxford and adumbrated his Rules in 1976 while at the Massachusetts Institute of Technology. He has studied biosynthesis (the way living things make molecules) extensively, especially in relation to the penicillins, and has applied many biosynthetic ideas to laboratory synthetic problems. Baldwin's rules differ in a fundamental way from the Woodward–Hoffmann rules you will meet in Chapters 34 and 35.

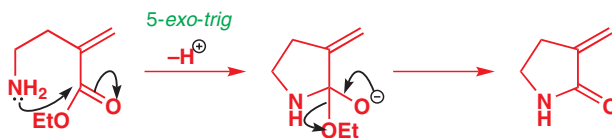
The Woodward–Hoffmann rules were deduced from theory, and examples were gradually discovered that fitted them. They cannot be violated: a 'forbidden' reaction that appears to disobey the Woodward–Hoffmann rules is getting around them by following a different mechanism. Baldwin's rules were formulated by making observations of reactions that do, or do not, work. This is why they are couched in terms of 'favoured' and 'disfavoured' reactions, rather than 'allowed' and 'forbidden' ones.

■ This is a crossover experiment. See Chapter 39, p. 1038.

■ Amines are usually good at undergoing conjugate addition to unsaturated esters: see Chapter 22.

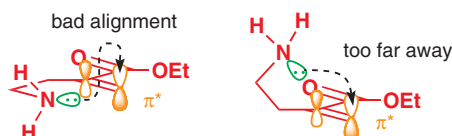


But this reaction doesn't happen: instead, the amine attacks the carbonyl group in a (favoured) 5-exo-trig cyclization.

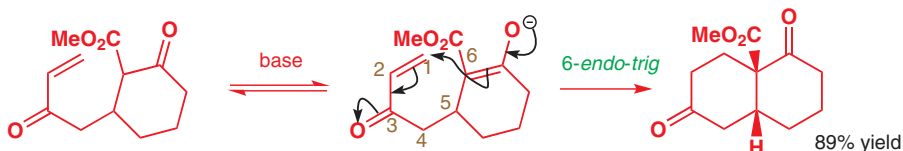


Interactive mechanism for 5-endo-trig vs 5-exo-trig

■ It's easier to see this with a model, and if you have a set of molecular models you should make one to see for yourself.



Lengthen the chain, though, and the dog gets his dinner. Here's a perfectly straightforward 6-endo-trig, for which orbital overlap presents no problem.

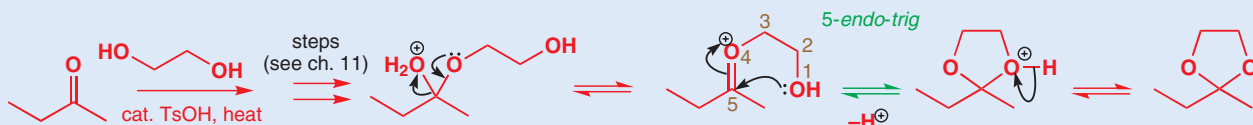


Interactive mechanism for 6-endo-trig cyclization

Exceptions to Baldwin's rules

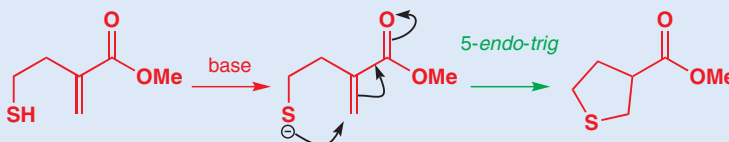
Baldwin's rules are only guidelines and, when a reaction is thermodynamically very favourable (Baldwin's rules, of course, describe the *kinetic* favourability of a reaction) and there is no other possible pathway, 5-endo-trig reactions *can* take place. The most striking example is one that you met quite early on in this book (Chapter

11): the formation of a cyclic acetal (dioxolane) from a carbonyl compound and ethylene glycol. We don't need to give again the full mechanism here, but you should check that you can still write it. The key step with regard to Baldwin's rules is shown with a green arrow. It's a 5-endo-trig reaction but it works!



In fact, cations frequently disobey Baldwin's rules. Other well-defined exceptions to Baldwin's rules include pericyclic reactions (Chapters 34 and 35) and reactions in which second-row atoms such as sulfur are included in the ring.

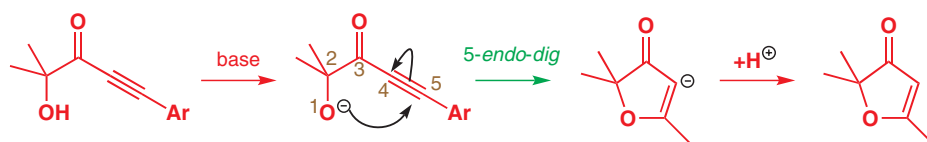
This 5-endo-trig reaction, the sulfur analogue of the amine cyclization that didn't work, is fine. C–S bonds are long, and the empty 3d orbitals of sulfur may play a role by providing an initial interaction with the C–C π orbital.



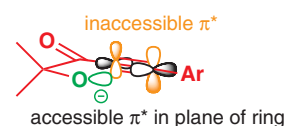
With *tet* and *trig* cyclizations, *exo* is better than *endo*; with *dig* cyclizations, the reverse is true.

● All *endo-dig* cyclizations are favoured.

Move from 5-*endo-trig* to 5-*endo-dig*, and the reactions become much easier: even 4-*endo-dig* reactions work. Here is an example of 5-*endo-dig*.



We warned you to look out for 5-*endo-trig* reactions because they are disfavoured even though on paper they look fine. Now the alert is the other way round! We expect you'd agree that these *endo-dig* reactions look awful on paper: the linear alkyne seems to put the electrophilic carbon well out of reach of the nucleophile, even further away than in the 5-*endo-trig* reaction. The important thing with *endo-dig* cyclizations, though, is that the alkyne has two π^* orbitals, one of which must always lie in the plane of the new ring, making it much easier for the nucleophile to get at.

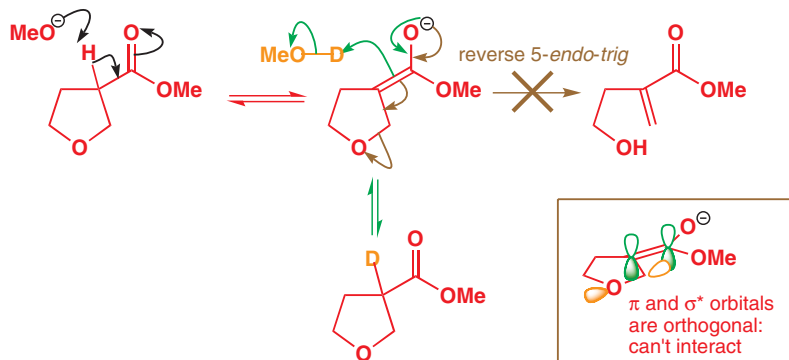


● 3 and 4-*exo-dig* are disfavoured; 5 to 7-*exo-dig* are favoured.

These reactions are less important and we will not discuss them in detail.

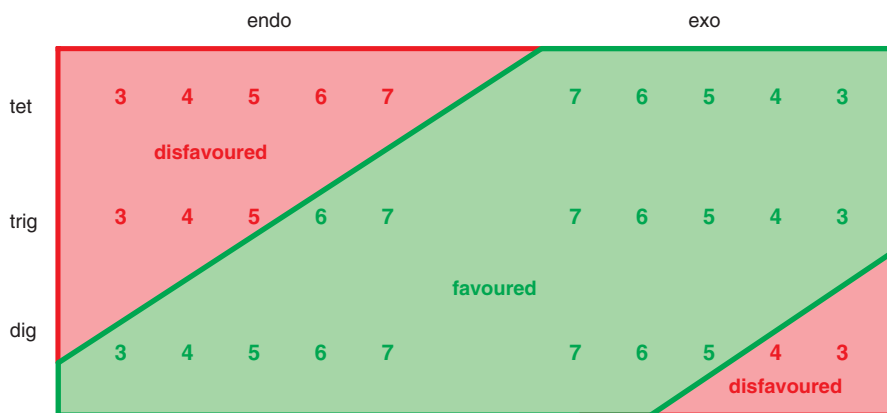
Baldwin's rules and ring opening

Baldwin's rules work because they are based on whether or not orbital overlap can be readily achieved in the conformation required at the transition state. But the transition state is the same whether the reaction is going forwards or backwards—the *principle of microscopic reversibility* (which is discussed further in Chapter 39) says that, if a reaction goes via a certain mechanism, the reverse reaction must follow exactly the same path in the opposite direction. So Baldwin's rules also apply to ring-opening reactions. This is where the unfavourability of 5-*endo-trig* really is important: this tetrahydrofuran-2-yl ester, for example, looks set up to do an E1cB elimination in base. Indeed, when it is treated with methoxide in deuterated methanol it exchanges the proton α to the ester for deuterium, proving that the enolate forms. But it does not eliminate: elimination would be a reverse 5-*endo-trig* process and is disfavoured.



Whenever you think about a ring-opening reaction, consider its reverse, and assess whether it is favoured according to Baldwin's rules.

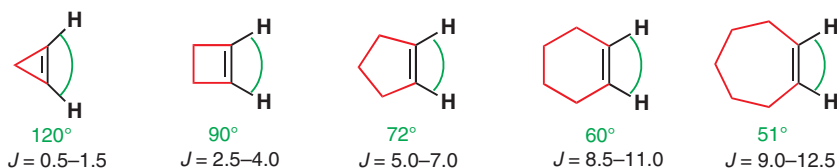
Baldwin's rules can be summarized in a chart. You should note the general outline of this chart: commit to memory that, broadly speaking, *endo-tet* and *endo-trig* are disfavoured; *exo-tet* and *exo-trig* are favoured, and the reverse for *dig*. Then you just need to learn the cut-off points that indicate the exceptions to this broad-brush view: 6-*endo-trig* falls into the favoured category while 4-*exo-dig* falls into the disfavoured one. And, if you really can remember only one thing, it should be that 5-*endo-trig* is disfavoured!



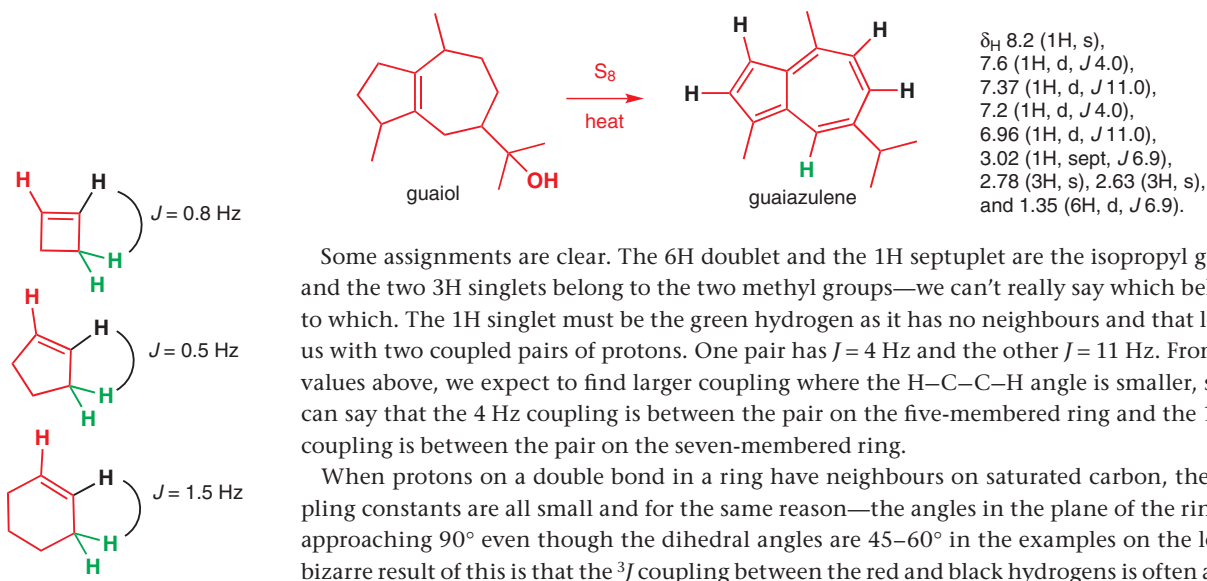
Ring size and NMR

On p. 796 we considered the effect of changes in the dihedral angle on coupling constants. But the dihedral angle is not the only angle worth measuring: we should also consider how the two C–H bonds are spread out in space. The dihedral angle is what we see when we look down the spine of the book in our earlier analogy—now we want to look at the pages in the normal way, at right angles to the spine, as if we were going to read the book. We can show what we mean by fixing the dihedral angle at 0° (the C–H bonds are in the same plane) and looking at the variation of J with the ring size of some simple cyclic alkenes.

■ The angles shown are calculated assuming the structures are regular planar polygons, and the coupling constants J are given in Hz.



The wider apart the hydrogens are spread, the smaller the coupling constant. Remember, the dihedral angle stays the same (0°)—we are just varying the angle in the plane. A dramatic illustration of this comes with the product of dehydrogenation of the natural product guaiol with elemental sulfur. From the brown, smelly reaction mixture, guaiazulene, a deep blue oil, can be distilled.



Some assignments are clear. The 6H doublet and the 1H septuplet are the isopropyl group, and the two 3H singlets belong to the two methyl groups—we can't really say which belongs to which. The 1H singlet must be the green hydrogen as it has no neighbours and that leaves us with two coupled pairs of protons. One pair has $J = 4$ Hz and the other $J = 11$ Hz. From the values above, we expect to find larger coupling where the H–C–C–H angle is smaller, so we can say that the 4 Hz coupling is between the pair on the five-membered ring and the 11 Hz coupling is between the pair on the seven-membered ring.

When protons on a double bond in a ring have neighbours on saturated carbon, the coupling constants are all small and for the same reason—the angles in the plane of the ring are approaching 90° even though the dihedral angles are 45 – 60° in the examples on the left. A bizarre result of this is that the 3J coupling between the red and black hydrogens is often about

the same as the allylic (4J) coupling between the red and the green hydrogens. An example follows in a moment.

The ‘spreading out’ effect also affects vicinal (3J) couplings in simple saturated rings. No other ring size has so well defined a conformation as that of the six-membered ring, but we can still note useful trends as we move from 6 to 5 to 4 to 3. Briefly, in five-membered rings, *cis* and *trans* couplings are about the same. In four- and three-membered rings, *cis* couplings are larger than *trans*. But in all cases the absolute values of J go down as the ring gets smaller and the C–H bonds are ‘spread out’ more. Indeed, you can say that all coupling constants are smaller in small rings, as we shall see. But we need to examine a few examples in a bit more detail.

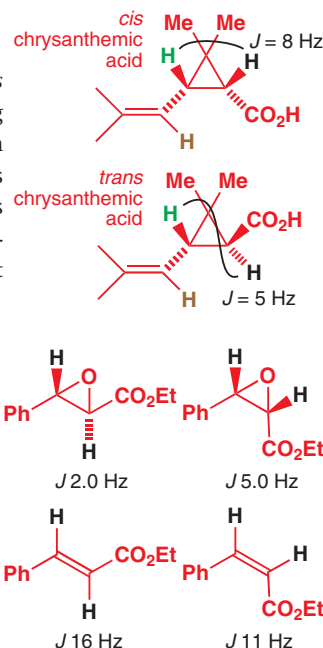
Three-membered rings

Three-membered rings have to be flat with all bonds eclipsed so the dihedral angle is 0° for *cis* Hs and 109° for *trans* Hs. Looking at the Karplus curve on p. 796, we expect the *cis* coupling to be larger, and it is. A good example is chrysanthemic acid, which is part of the pyrethrin group of insecticides found in the pyrethrum plant. Both *cis* and *trans* chrysanthemic acids are important, and in both isomers the coupling between the green proton on the ring and its brown neighbour on the double bond is 8 Hz. In the *cis* compound, the green proton is a triplet so the *cis* coupling in the ring is also 8 Hz. In the *trans* compound it is a double doublet with the second coupling, *trans* across the ring to the black H, of 5 Hz.

The most important three-membered rings are the epoxides. You saw in Chapter 13 (p. 295) that electronegative atoms reduce coupling constants by withdrawing electron density from the bonds that transmit the coupling ‘information’. This means that epoxide couplings are very small—much smaller than those of their closely related alkenes, for example. Compare the four coupling constants in the diagram: for the epoxide, all couplings are small, but *cis* coupling is larger than *trans* coupling. In alkenes, *trans* coupling is larger (Chapter 13, p. 293). The table summarizes the coupling constants for alkenes, epoxides, and cyclopropanes.

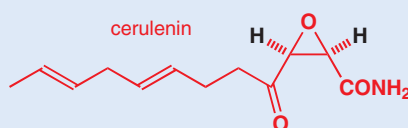
Typical coupling constants J , Hz

Stereochemistry	Alkene	Cyclopropane	Epoxide
<i>cis</i>	10–12	8	5
<i>trans</i>	14–18	5	2

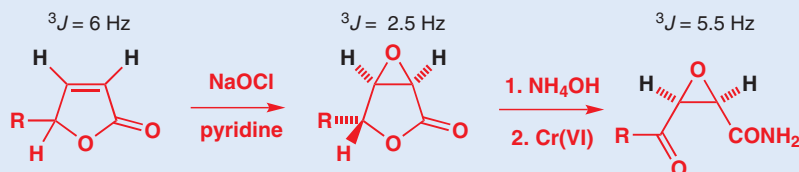


■ The epoxides have much smaller coupling constants than the alkenes because (1) the C–C bond is longer than the C=C bond, (2) there is an electronegative element, and (3) the ‘spreading out’ effect of the small ring comes into play.

Cerulenin

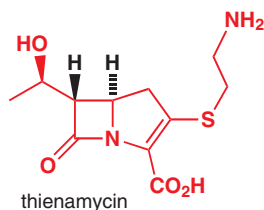
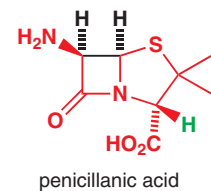


The natural product cerulenin is an antibiotic containing a *cis* epoxide. The coupling constant between the black hydrogens is 5.5 Hz. The compound has been made from an unsaturated lactone by epoxidation and ring opening. Follow what happens to the coupling constant between the black hydrogens as this sequence develops.



The *cis* coupling in the alkene is small because it is in a five-membered ring. It gets smaller in the bicyclic epoxide because the black Hs are now in both five- and three-membered rings and both are next to oxygen, but it gets larger in cerulenin itself because the five-membered ring has been opened.

Four-membered rings

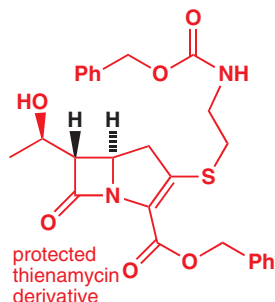


A similar situation exists with four-membered rings—the *cis* coupling is larger than the *trans* but they are generally both smaller than those in larger rings. A good example is the amino acid in the margin, the skeleton of the penicillins. The NMR spectrum contains three 1H signals in the middle regions. There is a singlet at δ_{H} 4.15 ppm that clearly belongs to the isolated green proton and two doublets at δ_{H} 4.55 and 5.40 ppm that must belong to the black protons. The coupling constant between them is 5 Hz and they are *cis*-related.

There are now large numbers of β -lactam antibiotics known and one family has the opposite (*trans*) stereochemistry around the four-membered ring. The typical member is thienamycin. We will analyse the spectrum in a moment, but first look at the differences—apart from stereochemistry—between this structure and the last. The sulfur atom is now outside the five-membered ring, the acid group is on a double bond in the same ring, and the amino group has gone from the β -lactam to be replaced by a hydroxyalkyl side chain.

Turning to the spectrum and the key question of stereochemistry, this is what the Merck discoverers said in their original article: ‘¹H NMR spectra of thienamycin (and derivatives)... show small vicinal coupling constants $J \leq 3$ Hz for the two β -lactam hydrogens. Past experience with penicillins... shows the *cis* relationship of the β -lactam hydrogens to be always associated with the larger coupling.’ As we have just seen penicillins have $J \sim 5$ Hz for these hydrogens.

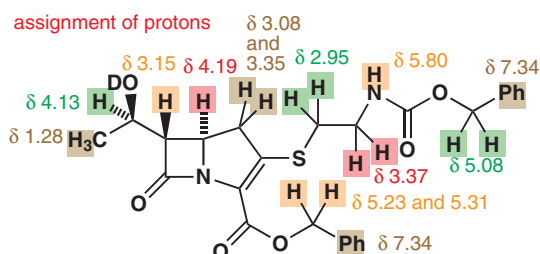
The NMR spectrum of a thienamycin derivative with protecting groups on the amine and carboxylic acids is shown below. Try your hand at interpreting it before you read the explanation. Your aim is to find the coupling constant across the four-membered ring.

NMR spectrum of thienamycin derivative in CD₃OD

Shift (δ_{H}), ppm	Integration	Multiplicity	Coupling constants (J), Hz
1.28	3H	d	6.5
2.95	2H	m	not resolved
3.08	1H	dd	9, 18
3.15	1H	dd	2.5, 7
3.35	1H	dd	9, 18
3.37	2H	m	not resolved
4.13	1H	dq	7, 6.5
4.19	1H	dt	2.5, 9
5.08	2H	s	—
5.23 and 5.31	2H	AB system ^a	AB system: ^a 12.5
5.80	1H	broad	—
7.34	10 H	multiplet	not resolved

^aSee p. 297 for discussion of AB systems.

The simple answer is 2.5 Hz. The signals at 3.15 and 4.19 ppm are the protons on the β -lactam ring and the 9 Hz extra coupling is to the CH₂ in the five-membered ring. If you went into this spectrum in detail you may have been worried about the 12.5 and especially the 18 Hz couplings. These are ² J (geminal) couplings and we will discuss them in the next section. The full assignment is shown below.



We should emphasize that a coupling constant of 5 or 2.5 Hz in isolation would not allow us to assign stereochemistry across the four-membered ring but, when we have both, we can say with confidence that the larger coupling is between *cis* Hs and the smaller coupling is between *trans* Hs.

Five-membered rings

You can visualize the conformation of a five-membered ring simply as a chair cyclohexane with one of the atoms deleted. But this picture is simplistic because the five-membered ring flexes (rather than flips) and any of the carbon atoms can be the one out of the plane. All the hydrogen atoms are changing positions rapidly and the NMR spectrum ‘sees’ a time-averaged result. Commonly, both *cis* and *trans* couplings are about 8–9 Hz in this ring size.

The best illustration of the similarity of *cis* and *trans* couplings in five-membered rings is a structure that was incorrectly deduced for that very reason. Canadensolide is an antifungal compound found in a *Penicillium* mould. The gross structure was quite easy to deduce from the mass spectrum, which gave the formula $C_{11}H_{14}O_4$ by exact mass determination, the infrared, which showed (at 1780 and 1667 cm^{-1}) a conjugated five-ring lactone, and some aspects of the proton NMR. The proposed structure is shown in the margin.

The stereochemistry of the ring junction Hs (shown in black and green) is not in question. They are certain to be *cis* as it is virtually impossible for two five-membered rings to be fused *trans*. The stereochemical uncertainty involves the third stereogenic centre on the left-hand ring. The coupling constant between the black and green Hs is 6.8 Hz, while that between the green and brown Hs is 4.5. Is this different enough for them to be *trans*? The original investigators decided that it was.

The mistake emerged when some Japanese chemists made this compound by an unambiguous route. The NMR spectrum was quite like that of canadensolide, but not the same. In particular, the coupling between the green and brown Hs was 1.5 Hz—quite different! So they also made the other possible diastereoisomer and found that it was identical to natural canadensolide. The details are in the margin.

An example of vicinal coupling in structural analysis: aflatoxins

We can bring together a lot of these points in the structure of one compound, the dreaded aflatoxin. Aflatoxins were mentioned in Chapter 19: they occur in moulds, including those that grow on some foods, and cause liver cancer. These slow-acting poisons are among the most toxic compounds known. Aflatoxin B_1 is an example. The four red protons on saturated carbons in the five-membered ring in the margin appear as two triplets: δ_H 2.61 (2H, t, J 5 Hz) and δ_H 3.42 (2H, t, J 5 Hz). The *cis* and *trans* couplings are the same. The yellow proton, on the junction between the two five-membered cyclic ethers, is a doublet δ_H 6.89 (1H, d, J 7 Hz). This is, of course, the *cis* coupling to the black hydrogen. The black hydrogen has this coupling too, but it appears as a doublet of triplets with a triplet coupling of 2.5 Hz: δ_H 4.81 (1H, dt, J 7, 2.5, 2.5 Hz). These small couplings can only be to the two green hydrogens: the 3J and 4J couplings are indeed the same.

Finally there is another strange coincidence—each green hydrogen appears as a triplet with 2.5 Hz couplings. Evidently, the *cis* coupling across the double bond is also 2.5 Hz. We expect *cis* coupling in a cyclopentene to be small (it was 4 Hz in the azulene on p. 814), but not that small—it must be the electronegative oxygen atom that is reducing the value still further.

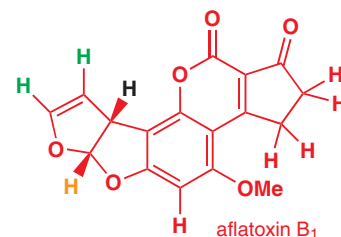
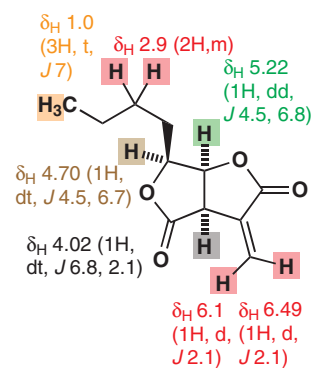
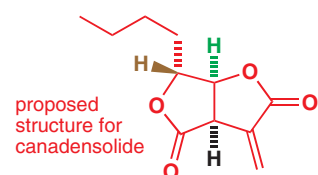
Geminal (2J) coupling

For coupling to be seen, the two hydrogen atoms in question must have different chemical shifts—identical protons do not couple. For 2J , or geminal, couplings the two hydrogen atoms are on the same carbon atom, so in order to discuss geminal coupling we must first consider what leads the two hydrogens of a CH_2 group to have different shifts.

To introduce the topic, an example. It may seem to you that any six-membered ring might show different chemical shifts for axial and equatorial groups. But this doesn't happen. Consider the result of this Robinson annelation reaction.

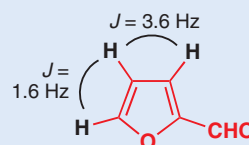


➔ This conformation is sometimes called an ‘envelope’. See Chapter 32, p. 834.

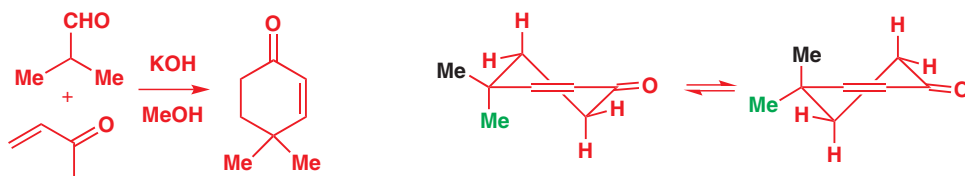


Coupling in furans

The size of coupling constants in five-membered rings containing oxygen is illustrated clearly in furfuraldehyde (furan-2-aldehyde): note how small the couplings are.

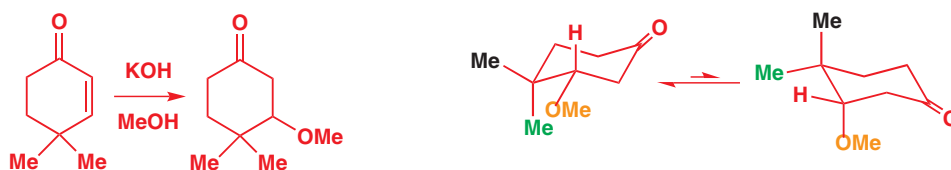


➡ The Robinson annelation was introduced in Chapter 26, p. 638. The 'flattened chair' conformation of cyclohexenones shown here is described in detail in Chapter 32, p. 830.



The two methyl groups at C4 give rise to a single signal in the ^{13}C NMR at 27.46 ppm. Even though one of them is (pseudo)axial and one (pseudo)equatorial, the molecule exists in solution as a rapidly equilibrating mixture of two conformations. The axial green methyl in the left-hand conformer becomes equatorial in the right-hand conformer, and vice versa for the black methyl group. The equilibrium position must be 50:50 and fast exchange averages the chemical shifts of the two methyl groups. The same is true for the CH_2 groups around the back of the ring, which each appear as a triplet.

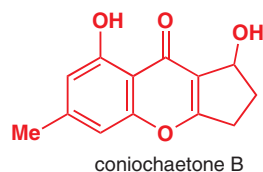
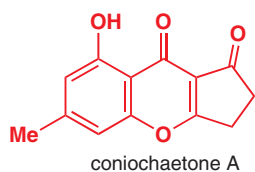
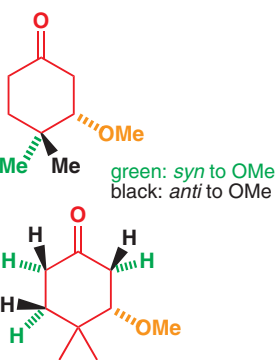
However, the enone is not the only product of this reaction. A methanol adduct is also formed by Michael addition of methanol to the conjugated enone. This product has two methyl signals at 26.1 and 34.7 ppm. If we examine the molecule by conformational analysis as we did for the first product we see a similar situation.



Similar but not the same. This time, the two conformations are not identical. One has the OMe group equatorial and the other has it axial. Even the two methyl groups do not entirely change places in the two conformations. True, the green methyl is axial on the left and equatorial on the right, but it has a gauche (dihedral angle 60°) relationship with the OMe group in *both* conformations. The black Me group is gauche to OMe on the left but anti-periplanar to the OMe group on the right. Averaging the two different conformations, in each of which the black and green methyl groups are different (that is, they don't just change places), does *not* lead to equalization of the two methyl groups.

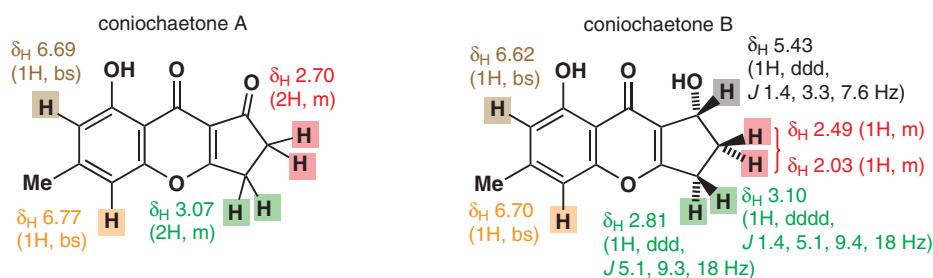
Perhaps a simpler way to discover this is to use a configurational, rather than a conformational, diagram. The green methyl group is on the same face of the molecule as the MeO group, while the black methyl group is on the other face. No amount of ring flipping can make them the same. They are *diastereotopic*, a term we shall define shortly. And so are all three CH_2 groups in the ring. The green Hs are on the same face of the molecule as the MeO group while the black Hs are on the other face.

A proton NMR example confirms this, and here is one from an odd source. There are fungi that live on animal dung, called coprophilous fungi. They produce antifungal compounds, presumably to fight off competition! Anyway, in 1995 two new antifungal compounds were discovered in a fungus living on lemming dung. They were named coniochaetones A and B and their structures were deduced with the usual array of mass and NMR spectra. The proton spectra, run on a 600 MHz machine, are shown below, and they reveal considerable detail.



Some of the spectrum is essentially the same for the two compounds, but other parts are quite different. Coniochaetone A has a very simple spectrum, very easily assigned. Coniochaetone B is rather more interesting. The spectrum is much more complicated, even though it has only one more C-H (the grey one) than coniochaetone A. The reason is that addition of that H atom creates a stereogenic centre and makes the top and bottom faces of the molecule different. Each H in both CH_2 groups becomes differentiated from its partner.

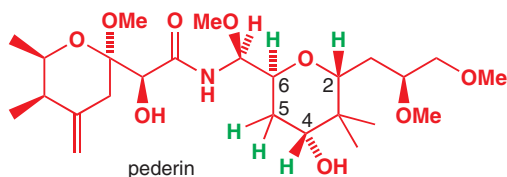
Coniochaetone A		Coniochaetone B	
δ_{H} , ppm	Coupling	δ_{H} , ppm	Coupling
2.41 (3H)	s	2.38 (3H)	s
2.70 (2H)	m	5.43 (1H)	ddd, J 1.4, 3.3, 7.6 Hz
3.07 (2H)	m	2.49 (1H)	m
6.77 (1H)	broad s	2.03 (1H)	m
6.69 (1H)	broad s	3.10 (1H)	dddd, J 1.4, 5.1, 9.4, 18 Hz
12.21 (1H) ^a	s	2.81 (1H)	ddd, J 5.1, 9.3, 18 Hz
		6.70 (1H)	broad s
		6.62 (1H)	broad s
		12.25 (1H) ^a	s

^aExchanges with D₂O.

The green Hs are coupled to each other ($J = 18$ Hz) and to each of the black Hs with a different coupling constant. One of the green hydrogens also shows a long-range ($^4J = 1.4$ Hz) W-coupling to the red H. The black Hs are too complex to analyse, even at 600 MHz, but the different couplings to the red Hs are shown by the signal at 5.43 ppm.

The size of the geminal coupling constant

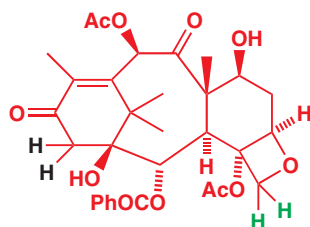
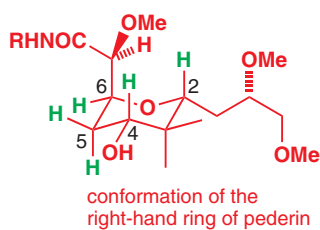
The 18 Hz geminal coupling constant between the green protons of coniochaetone B is large, but not unusually so for a geminal coupling. A more typical figure in a six-membered ring might be closer to 14 Hz, and we will see shortly why the value in coniochaetone B is bigger than this. The example below provides an opportunity to examine coupling constants in another example where NMR was essential for determining the structure. The compound is pederin, a toxic amide of the blister beetle *Paederus fuscipes*. After some incorrect early suggestions, the actual structure of the compound was eventually deduced as shown.



We are not going to discuss the full structure elucidation, but will concentrate on the stereochemistry of the right-hand ring. The five (green) protons on the ring gave the signals listed in the margin.

Three of the protons have shifts δ_{H} 3–4, and are obviously on carbons attached to oxygen atoms. The other two, δ_{H} about 2, must be the diastereotopic pair at C5. The coupling of 12 Hz, which appears in both signals, must be the geminal coupling and the other couplings are found in the signals at δ_{H} 3.75 and 3.85. The signal at δ_{H} 3.75 has no other couplings and must be from C4 so that leaves δ_{H} 3.85 for the hydrogen atom at C6, which is also coupled to the hydrogen in

δ_{H}
1.85 (1H, ddd, J 5, 10, 12)
2.10 (1H, ddd, J 3, 4, 12)
3.75 (1H, dd, J 4, 10)
3.85 (1H, ddd, J 3, 5, 8)
4.00 (1H dd, J 3, 7)



➡ The effect of electronegative atoms on coupling was discussed in Chapter 13, p. 295.

the side chain. The 10 Hz coupling must be axial–axial—the others are all much smaller, meaning there is just the one axial–axial coupling. The left-hand side chain must therefore occupy an axial position as shown in the margin. This is perhaps a bit surprising—it's large and branched—but the molecule has no choice but to place one of the two side chains axial.

One of the most important compounds from the last 25 years is Taxol, the anti-cancer compound isolated from the bark of the Pacific yew tree. Taxol's structure has four rings—with eight, six (twice), and four members—and is too complex to analyse in detail, but the NMR spectrum of the closely related compound in the margin gives us the opportunity to illustrate how much geminal couplings in rings may vary and to analyse some of the factors which control this variation. The coupling between the black Hs is 20 Hz while that between the green Hs is just 6 Hz.

20 Hz is a very large coupling constant, even for geminal coupling, and the reason it is so big is the adjacent π bond. If a CH_2 group is next to an alkene, aromatic ring, $\text{C}=\text{O}$ group, CN group, or any other π -bonded functional group, it will have a larger geminal coupling constant. This effect also explains the large 18 Hz coupling in coniochaetone B (p. 819).

But why is the green coupling so small? The reason is the four-membered ring. You saw on p. 814 that vicinal couplings are small in small rings; the same is true of geminal couplings. Another factor comes into play here as well—the adjacent oxygen atom. Electronegative atoms always tend to reduce coupling constants.

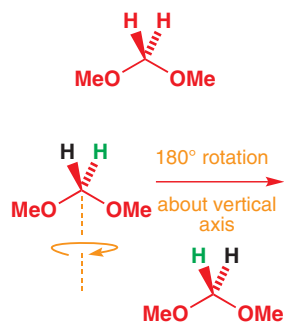
• The size of 2J and 3J coupling constants

We have now covered all of the important influences on the size of coupling constants. They are:

- dihedral angle: 3J greatest at 180° and 0° ; about 0 Hz at 90°
- ring size, which leads to 'spreading out' of bonds and lower 2J and lower 3J in small rings
- electronegative atoms, which decrease 2J and 3J coupling constants between protons
- π systems, which increase 2J coupling constants between protons.

Diastereotopic groups

■ To understand this discussion, it is very important that you understand the ideas that we covered in Chapter 14. You may need to refresh your memory of the stereochemical points there before you read further.



You have now seen several examples where two protons attached to the same carbon are not the same, and it is time to examine more closely the appearance of these CH_2 groups in NMR spectra. To do this, we shall have to discuss some aspects of symmetry that build on what you learned in Chapter 14. You will see that there are *three* possibilities for the symmetry associated with a CH_2 group, and these three possibilities have an effect both on the chemistry of the molecule and on what its NMR spectrum will look like.

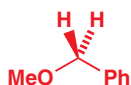
First, an example in which the two hydrogens are indeed the same. Although the molecule is of course achiral, we may draw one hydrogen coming towards us and one going away, but the two Hs are the same. This is easy to demonstrate. If we colour one H black and one green, and then rotate the molecule through 180° , the black H appears in the place of the green H and vice versa. The rotated molecule hasn't changed because the *other* two substituents (OMe here) are also the same.

If we had given out uncoloured models of this molecule with this book, and asked each reader to paint one H green and one H black, we would have no way at all of giving instructions about which to paint what colour. But it wouldn't matter because, even without these instructions, every reader would produce an identical model, whichever way they painted their Hs.

The correct description for this pair of hydrogen atoms is **homotopic**. They are the same (*homo*) topologically and cannot be distinguished by chemical reagents, enzymes, NMR machines, or human beings.

• Homotopic groups

Homotopic groups cannot be distinguished by any means whatsoever: they are chemically entirely identical.



What happens when the other two substituents are different? At first sight the situation does not seem to have changed. Surely the two hydrogens are still the same as one another?

In fact, they aren't—not quite. If we had given out uncoloured models of this molecule and just said 'paint one H green and one H black', we would not have got just one type of model. However, this time we *could* give instructions about which H we wanted which colour. To get the first of these two, we just need to say 'Take the MeO group in your *left* hand and the Ph group in your *right*, kink the carbon chain upwards. The hydrogen coming towards you is to be painted black.' All the models produced by readers would then be identical—as long as the readers knew their left from their right. This is a very important point: the green and black hydrogens in this molecule (unlike the first one) can be described only in phrases incorporating the words 'left' or 'right', and are distinguishable only by a system that knows its left from its right.

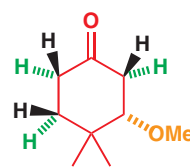
Human beings are such a system: so are enzymes and the asymmetric reagents you will meet in Chapter 41. But NMR machines are not. NMR machines cannot distinguish right and left—the NMR spectra of two enantiomers are identical, for example. There is no question of enantiomers in the molecule in question—it has a plane of symmetry and is achiral. Nonetheless, the relationship between these two hydrogens is rather like the relationship between enantiomers (the two possible ways of colouring the Hs are enantiomers—mirror images) and so they are called **enantiotopic**. Enantiotopic protons appear identical in the NMR spectrum.



● Enantiotopic groups

Enantiotopic groups can be distinguished by systems that can tell right from left, but are still magnetically equivalent and appear identical in the NMR spectrum.

The third situation usually arises when the molecule has a stereogenic centre. As an example we can take the Michael product from the beginning of this section. It is now very easy to distinguish the two hydrogens on each ring carbon atom and, if we want to give instructions on how to paint a model of this molecule, we can just say 'Make all the Hs on the same side of the ring as OMe green, and the ones on the opposite side to OMe black.' We do not need to use the words 'right' or 'left' in the instructions, and it is not necessary to know your right from your left to tell the two types of Hs apart. Ordinary chemical reagents and NMR machines can do it. These Hs are different in the way that diastereoisomers are different and they are **diastereotopic**. We expect them to have different chemical shifts in the proton NMR spectrum. The same is true of the methyl groups: they too are diastereotopic and we expect them to have different shifts.



● Diastereotopic groups

Diastereotopic groups are chemically different: they can be distinguished even by systems that cannot tell right from left, and they can appear at different chemical shifts in the NMR spectrum.

■ NMR machines *can* tell the difference, but it does not follow that they *will*. There are many examples of protons that are different but have the same chemical shift (toluene, PhMe, shows a singlet in the NMR for all its aromatic protons even though they are of three different kinds). Sometimes diastereotopic protons have the same chemical shift, sometimes slightly different chemical shifts, and sometimes very different chemical shifts.

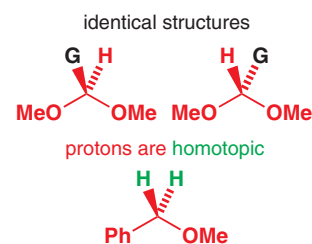
How to tell if protons are homotopic, enantiotopic, or diastereotopic

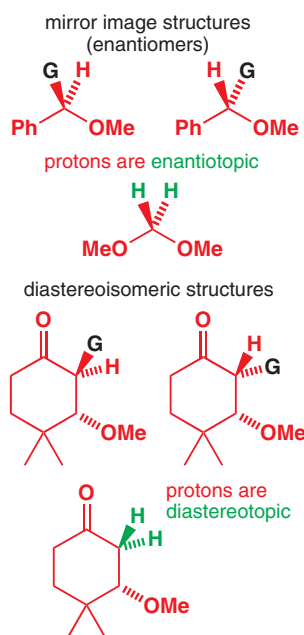
What we have said so far explains to you why homotopic and enantiotopic groups always appear identical in the NMR spectrum, but diastereotopic protons may not. Now we will give a quick guide to determining what sort of pair you are dealing with in a given molecule.

The key is to draw your molecule twice. In each drawing (or model if you prefer) replace one of the Hs (we'll assume we're looking at protons, but the argument works for other groups too—Me groups, for example) with an imaginary group 'G'. Write down the first structure you get, with stereochemistry shown. Next, write down the structure you get by replacing the other H with the group G. Now the more difficult bit: identify the stereochemical relationship between the two molecules you have drawn.

- If they are identical molecules, the protons are homotopic.
- If they are enantiomers, the protons are enantiotopic.
- If they are diastereoisomers, the protons are diastereotopic.

This is really just a simpler way of doing what we did with black and green above, but it is easy to do for any molecule. Take the first of our examples, and replace each H in turn by G. These two molecules are identical because just turning one over gives the other: the protons are homotopic.





■ The shape of NMR signals where J and the chemical shift difference are of the same order of magnitude were discussed in Chapter 13. The arguments apply to any coupled protons of similar chemical shift—there we used disubstituted aromatic rings as the example—but are particularly relevant here.

■ It is not always easy to decide which proton gives rise to which signal in a diastereotopic AB system, although the information may be important in assigning stereochemistry. The size of other couplings, or the nuclear Overhauser effect (p. 799), may assist.

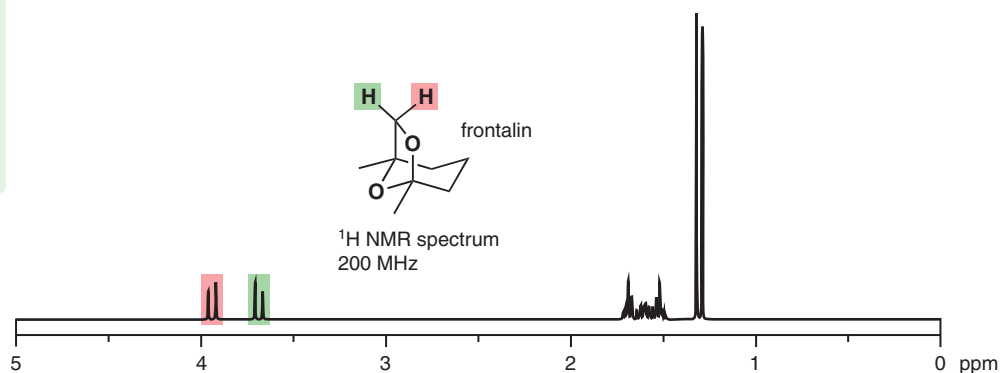
Now for the next example. The two molecules are not identical: to make one into the other you need to reflect in the plane of the paper, so they are enantiomers, and the Hs are *enantiotopic*. There is another term we must introduce you to in relation to this molecule, which will become useful in the next chapter, and that is 'prochiral'. The molecule we started with here was not chiral—it had a plane of symmetry. But by changing just one of the Hs to a different group we have made it chiral. Molecules that are achiral but can become chiral through one simple change are called **prochiral**.

Now we will choose one of the three pairs of Hs in the cyclohexanone example. The starting molecule is, of course, now chiral, and the two molecules we get when we replace each H by G are now diastereoisomers: one has G and OMe *anti*, the other *syn*, and the pairs of hydrogens are *diastereotopic*. The same is true for the other CH₂ groups. Furthermore, the methyl groups attached to the ring will be diastereotopic too, and we expect them to appear as two 3H singlets.

Spotting diastereotopic protons in the NMR spectrum

A CH₂ group with diastereotopic Hs isolated from any other Hs will give rise to two signals, one for each H, and they will couple to each other so that the complete signal is a pair of doublets. A typical geminal (² J) coupling constant is 14 Hz—relatively large. Because chemical shift differences ($\Delta\delta$) between Hs on the same carbon atom tend to be small—usually less than 1 ppm—the signals have $\Delta\delta \sim J$ and are distorted into a 'roof-topped' AB system.

Here is an example. The pheromone frontalin is a remarkable compound used by both insects and by elephants to attract a mate. Its structure and ¹H NMR spectrum are shown below.

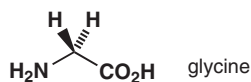


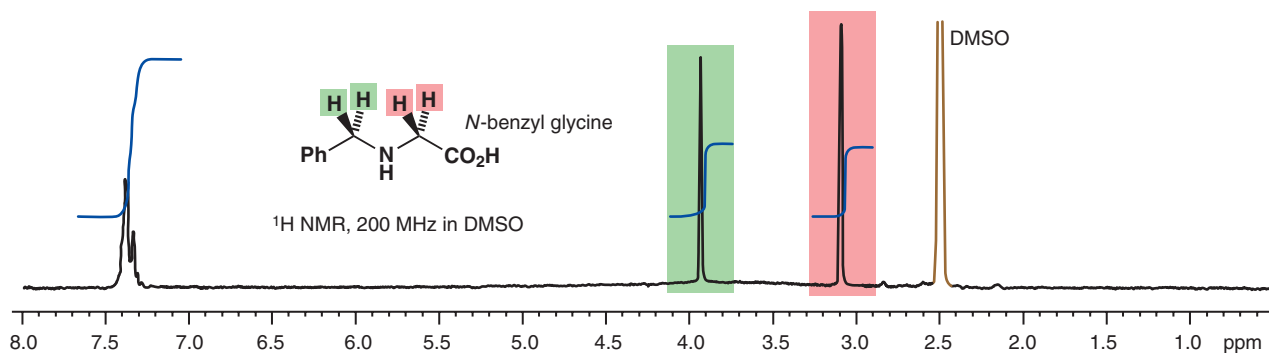
The red and green hydrogens are diastereotopic, and have no other couplings. They give the pair of doublets at 3.42 and 3.93 ppm., each with J 7 Hz (an AB system) in the ¹H NMR. The coupling constant here is small for ² J —only 7 Hz—but that should not surprise you since we have a five-membered ring and a nearby oxygen atom.

The coupling constant in an AB system is easy to extract—it is the difference in Hz between the two lines highlighted same colour in the spectrum above. But the chemical shifts are not so easily measured. The chemical shift of each proton is at the weighted mean of the two lines—the more distorted the signal, the nearer the chemical shift to that of the larger inner line.

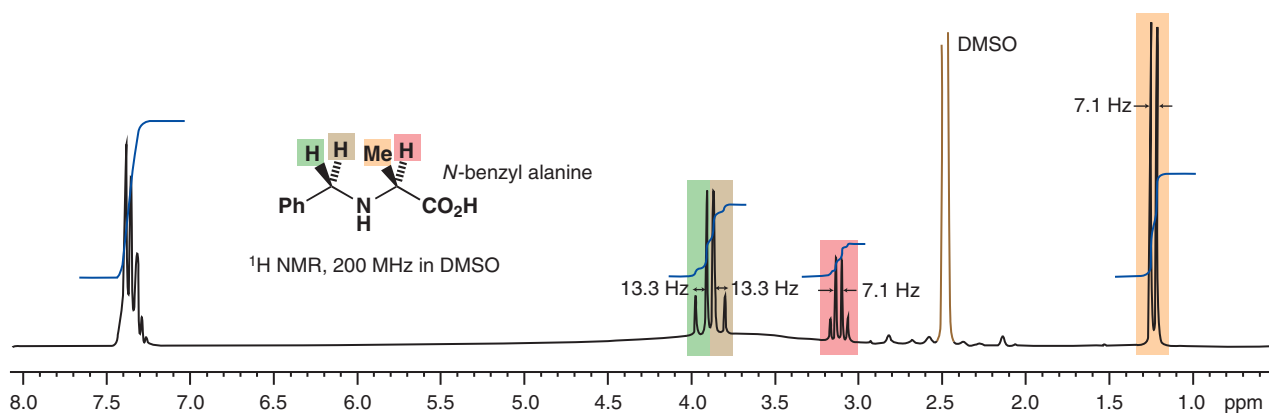
Diastereotopic protons in acyclic compounds

The same principles apply to open-chain compounds, such as amino acids. All of the amino acids in proteins except glycine are chiral. Glycine has a CH₂ group that gives a singlet in the NMR spectrum as its Hs are enantiotopic. Similarly, the *N*-benzyl derivative of glycine has a second CH₂ group (NCH₂Ph) that gives another singlet in the NMR spectrum as these Hs too are enantiotopic.





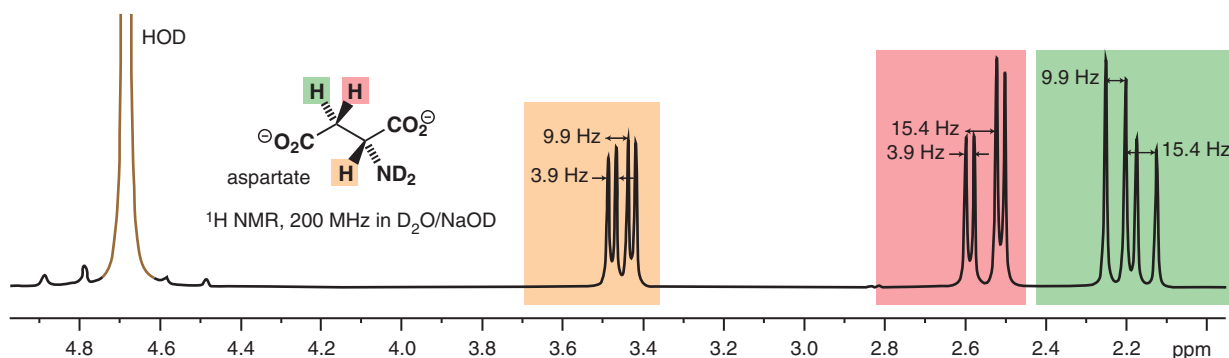
The plane of the paper is a plane of symmetry for both of the CH_2 groups of *N*-benzyl glycine in the way it is drawn here. But for the other amino acids, which are all chiral, the symmetry is different. The ^1H NMR spectrum of *N*-benzyl alanine is shown below. There is now no plane of symmetry, so the Hs of the NCH_2Ph group are diastereotopic. The CH_2 group appears as an AB pattern.



In the way in which the molecule is drawn, the green H is on the same side as the Me group and the brown H on the other. It does not matter that there is free rotation in this molecule—the two diastereotopic protons are never in the same environment so even after averaging over all the conformations available to the molecule they always appear at different chemical shift. If a molecule is chiral, all CH_2 groups in that molecule—however flexible it may be and however far they are from any chiral centre—are diastereotopic, and can potentially appear in the spectrum as an AB system.

It is more common to find diastereotopic CH_2 groups with neighbours, and an example arises when aspartic acid is dissolved in D_2O with NaOD present. The NH_2 protons are exchanged for deuterium atoms and do not show up in the spectrum—the molecule exists as its dianion.

■ For an illustration of diastereotopic CH_2 groups which may or may not appear as AB systems, look back at the spectrum of thienamycin on p. 816. Compare the two OCH_2Ph groups: both have a diastereotopic CH_2 pair, but one appears as a singlet and one as an AB system.



To summarize...

We have covered a lot of ground in this chapter, and have used the huge topic of saturated heterocycles to explain a lot, not just about the reactivity and conformation of rings. Many of these explanations involved consideration of the alignment of orbitals—we called these stereoelectronic effects. The same analysis allowed us to make sense of the NMR spectra, and in particular the coupling constants, of cyclic molecules, both heterocyclic and carbocyclic. And by thinking about symmetry in these cyclic molecules we were also able to deduce the origins of symmetry-related features (such as diastereotopic protons) in the NMR spectra of acyclic compounds.

The next chapter is the fourth consecutive chapter to take rings as a theme. It will introduce you to practical ways of controlling stereochemistry in cyclic systems—the first step towards making molecules with a particular stereochemistry, which will continue in Chapter 33 and culminate in Chapter 41 on asymmetric synthesis.

Further reading

Another reminder: you will find it an advantage to have one of the short books on spectroscopic analysis to hand as they give explanations, comprehensive tables of data, and problems. We recommend D. H. Williams and Ian Fleming, *Spectroscopic Methods in Organic Chemistry*, McGraw-Hill, London, 6th edn, 2007.

For stereoelectronics, a short introduction is A. J. Kirby, *Stereoelectronic effects*, OUP, Oxford, 1996, and a longer book is P. Deslongchamps, *Stereoelectronic Effects in Organic Chemistry*, Pergamon, London, 1983.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>